The maximal energy of integral circulant graphs of orders $p^{2r}3^{2s+1}$, pq^m and p^2q^{b*}

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Abstract

Jiang and Yang determined the maximal energy of integral circulant graphs of order $n = p^{2r}q^{2s+1}$, where p, q are distinct odd primes with $q \geq 5$: the checkerboard divisor set $\{p^iq^j : i+j \text{ even}\}$ is the unique maximiser. They noted that their semidefinite method does not extend to $q = 3$. We prove the remaining case: for every prime $p \geq 5$ and all $r \geq 1$, $s \geq 0$, the checkerboard set is the unique energy maximiser for order $p^{2r}3^{2s+1}$, with the maximal energy given by the formula of Jiang and Yang. This settles the maximal-energy problem for opposite-parity exponents for all pairs of distinct odd primes. The key step is a sharp inequality for sign matrices under weighted Ramanujan transforms. We prove it with a path identity and an explicit potential function, and the same argument gives a semidefinite-free proof of the theorem of Jiang and Yang. For exponents of equal parity the checkerboard set contains n and is not admissible. We determine the maximal energy, and its two maximisers, when one of the exponents is 1 or 2, for all distinct odd primes p, q, including 3, the other exponent being arbitrary; for $n = pq$ it also follows from known energy formulas. For example, for $n = p^2q^b$ with b even the maximal energy is $\frac{1}{2}[n + (5p^2 - 8p + 4)\, d_b(q)] - (p^2 - 2p + 2)\, \delta_b(q)$, where d_b and δ_b are explicit polynomials, and it is attained exactly by $\{p^iq^j : i+j \text{ odd}\}$ and $\{p^iq^j : i+j \text{ even}\} \setminus \{n\}$. With the opposite-parity case, this determines the maximal energy for every order p^aq^b with $\min\{a, b\} \leq 2$. Version 1.5 of this paper also claimed the range $3 \leq \min\{a, b\} \leq 8$ by a computer-assisted argument; version 2 withdraws that claim, because it has no formal proof, and keeps only a conditional reduction (Proposition 31). Jiang and Yang [7] have since stated the corresponding formula, $\frac{1}{2}[n + d_a(p)d_b(q)] - \delta_a(p)\delta_b(q)$, and the same description of the two maximisers, for all exponents of equal parity and distinct primes $p, q \geq 5$; their statement does not include the prime 3. All the mathematics of this paper, including the closed forms of d_k and δ_k, is formally verified in Lean 4; statements about the literature are not. The proofs were found by AI coding agents working under the author's direction; their role is described at the end of the paper. No human mathematician has reviewed this paper.

[*] *Version 2.* Every result of this version is formally verified in Lean 4 (Section 13); version 1.5, the previous one, was only partly formalised. Version 2 narrows the results on exponents of equal parity: Theorem 5 is now stated for $\min\{a, b\} \leq 2$, the other exponent being arbitrary, instead of $\min\{a, b\} \leq 8$. The case $3 \leq \min\{a, b\} \leq 8$ of that theorem and Proposition 15 of version 1.5 rested on computer calculations and have no formal proof, so they are withdrawn; the withdrawal does not report an error in those calculations. Other claims without a formal proof, mainly the computational cross-checks of version 1.5, are deleted. The numbers of the remaining statements are unchanged, and the number 15 is not used. The main changes are listed at the end of Section 13. Earlier versions remain available in the releases of [1]: version 1.5 in 1.5.0, version 1.4 in 1.4.0 and version 1.2 in 1.2.0. Theorem 10, Conjecture 12 and Proposition 13 of version 1.2 are Theorem 12, Conjecture 14 and the withdrawn Proposition 15.

1 Introduction

The energy of a finite graph is the sum of the absolute values of its adjacency eigenvalues [2]. For $n \geq 2$ and a nonempty set $\mathcal{D}$ of proper divisors of n, the integral circulant graph $\mathrm{ICG}_n(\mathcal{D})$ has vertex set $\mathbb{Z}/n\mathbb{Z}$, two distinct vertices u, v being adjacent when $\gcd(u-v, n) \in \mathcal{D}$. These are the circulant graphs with integral spectrum [13]. The problem of maximising the energy of $\mathrm{ICG}_n(\mathcal{D})$ over $\mathcal{D}$ for fixed n has been solved for prime powers [11] and studied for other orders; see the introduction of [6] for a survey. Roldán [10] conjectured that for $n = p^2q^3$ the checkerboard set $\{1, p^2, pq, q^2, p^2q^2, pq^3\}$ is the unique maximiser. Jiang and Yang [6] proved much more.

Theorem 1 (Jiang–Yang [6, Theorem A])**.** *Let p, q be distinct odd primes with $q \geq 5$, let $r \geq 1$, $s \geq 0$ and $n = p^{2r}q^{2s+1}$. Among all nonempty sets of proper divisors of n, the energy of $\mathrm{ICG}_n(\mathcal{D})$ is uniquely maximised by*

$$\mathcal{D}^*_{r,s} = \{p^i q^j : \ 0 \leq i \leq 2r, \ 0 \leq j \leq 2s+1, \ i+j \ \textit{even}\},$$

and the maximal energy is $\frac{1}{2}\left[n + d_{2r}(p)\, d_{2s+1}(q)\right]$, where $d_k(x) = (2k+1)x^k + 4\sum_{j=0}^{k-1}(-1)^{k-j}(j+1)x^j$.

They remark that the case $q = 3$ is not covered, because their odd-exponent semidefinite argument fails at $x = 3$ [6, Remark 4.2]. Our main result removes this restriction.

Theorem 2. *The conclusion of Theorem 1 holds for $q = 3$: for every prime $p \geq 5$ and all $r \geq 1$, $s \geq 0$, the set $\mathcal{D}^*_{r,s}$ is the unique energy maximiser among nonempty sets of proper divisors of $n = p^{2r}3^{2s+1}$, and the maximal energy is $\frac{1}{2}\left[n + d_{2r}(p)\, d_{2s+1}(3)\right]$.*

Together with Theorem 1 (which allows $p = 3$ on the even side), Theorem 2 determines the maximal energy and the unique maximiser for every order $p^{\text{even}}q^{\text{odd}}$ with p, q distinct odd primes. The case $r = s = 1$ was treated earlier. Park [9] claims a proof of Roldán's conjecture for all distinct odd primes. Schreib [12] gives a written proof of the case $q = 3$ and combines it with [6] to obtain the conjecture. An arithmetic certificate for $n = 27p^2$ was announced in [14]. These works concern only the exponents $(2, 3)$, and, when Theorem 2 was released in version 1.1.0 of [1] on 26 September 2026, we were not aware of any earlier treatment of the case $q = 3$ for general r and s.

If the two exponents have the same parity, the checkerboard set contains n itself and is not a set of proper divisors, and the maximiser changes. Conjecture 14, stated in earlier versions of this paper, gives two maximisers and an explicit maximal energy for this case. We prove it when one of the exponents equals 1 or 2. For $n = p^a q^b$, a set written as $\{p^i q^j : \ \ldots\}$ consists of the divisors $p^i q^j$ of n ($0 \leq i \leq a$, $0 \leq j \leq b$) with the stated property, and $\delta_k(x) = (x^k(x-1) + 2(-1)^k)/(x+1)$ (Remark 9).

Theorem 3. *Let p, q be distinct odd primes, let $m \geq 1$ be odd, and let $n = pq^m$. The maximal energy of an integral circulant graph of order n is*

$$\tfrac{1}{2}\left[n + (3p-4)\, d_m(q)\right] - (p-2)\, \delta_m(q),$$

and it is attained exactly by the sets $\{p^i q^j : \ i+j \ \textit{odd}\}$ and $\{p^i q^j : \ i+j \ \textit{even}\} \setminus \{n\}$. Exchanging the names of the primes, for $n = p^m q$ the maximal energy is $\frac{1}{2}\left[n + (3q-4)\, d_m(p)\right] - (q-2)\, \delta_m(p)$, attained exactly by the two sets with the same description.

For $m = 1$ the maximal energy is $4(p-1)(q-1)$, attained by $\{p, q\}$ and $\{1\}$. This case can also be derived from published energy formulas of Ilić [3] and of Ilić and Bašić [4] (Remark 16), although we have not found it stated. For the exponent 2 we prove the following.

Theorem 4. *Let p, q be distinct odd primes, let $b \geq 2$ be even, and let $n = p^2 q^b$. The maximal energy of an integral circulant graph of order n is*

$$\tfrac{1}{2}\left[n + (5p^2 - 8p + 4)\, d_b(q)\right] - (p^2 - 2p + 2)\, \delta_b(q),$$

and it is attained exactly by the sets $\{p^i q^j : i + j \text{ odd}\}$ and $\{p^i q^j : i + j \text{ even}\} \setminus \{n\}$. Exchanging the names of the primes, for $n = p^b q^2$ the maximal energy is $\frac{1}{2}\left[n + (5q^2 - 8q + 4)\, d_b(p)\right] - (q^2 - 2q + 2)\, \delta_b(p)$, attained exactly by the two sets with the same description.

Here $5x^2 - 8x + 4 = d_2(x)$ and $x^2 - 2x + 2 = \delta_2(x)$. The proof of Theorem 4 gives a sign-matrix inequality for all real $p \geq 5$, $q \geq 3$ and for $p = 3$, $q \geq 5$ (Theorem 25). Since also $3x - 4 = d_1(x)$ and $x - 2 = \delta_1(x)$, Theorems 3 and 4 together give the following.

Theorem 5. *Let p, q be distinct odd primes, let $a, b \geq 1$ with $a + b$ even and $\min\{a, b\} \leq 2$, and let $n = p^a q^b$. The maximal energy of an integral circulant graph of order n is $\frac{1}{2}\left[n + d_a(p)\, d_b(q)\right] - \delta_a(p)\, \delta_b(q)$, and it is attained exactly by the sets $\{p^i q^j : i + j \text{ odd}\}$ and $\{p^i q^j : i + j \text{ even}\} \setminus \{n\}$.*

This is Theorem 3 if $\min\{a, b\} = 1$ and Theorem 4 if $\min\{a, b\} = 2$; the other exponent is arbitrary, and the prime 3 is allowed. Version 1.5 stated this theorem for $\min\{a, b\} \leq 8$; see the footnote on the title page.

Theorems 1 and 2 cover every order whose exponents have opposite parity. So Theorems 1–5 determine the maximal energy, and all maximisers, for every order $p^a q^b$ with $\min\{a, b\} \leq 2$, and for every order whose exponents have opposite parity, where p, q are distinct odd primes.

Earlier searches. Our searches for earlier work on exponents of equal parity were made on 26 September 2026, between 03:37 and 04:32 UTC, in arXiv, OpenAlex, Semantic Scholar, Google Scholar, web search engines, and GitHub code, issues, pull requests and commits, including the repositories of [9, 12, 14]; an incremental search of arXiv, OpenAlex and GitHub followed between 05:53 and 05:58 UTC. They found no earlier determination of the maximal energy for such orders; for $n = pq$ it can be derived from known energy formulas (Remark 16). A further check on 26 September 2026 between 15:49 and 15:52 UTC, repeated for version 1.5 between 19:04 and 19:06 UTC (the arXiv, including the versions of [6]; GitHub, including the issues of [14]; MathDB; and MathOverflow, through the Stack Exchange API) found nothing new. This reports the coverage of those searches at that time, not a guarantee of priority.

Later work. After version 1.5 was released, Jiang and Yang [7] (27 October 2026) stated in their Theorem A that for distinct primes $p, q \geq 5$ and $a, b \geq 1$ with $a + b$ even, the maximal energy of an integral circulant graph of order $n = p^a q^b$ is $\frac{1}{2}[n + d_a(p) d_b(q) - 2 g_a(p) g_b(q)]$, attained exactly by the two sets of Theorem 5; their g_k is our δ_k. So for $p, q \geq 5$ Theorem 5 is a special case of their statement, while their statement does not include the prime 3. They add that, together with [6], this determines the maximal energy and all maximisers for every order $p^a q^b$ with distinct primes $p, q \geq 5$. We have compared the statements only. We have not reviewed their proof, and nothing in this paper or in its Lean proofs uses it. The orders of equal parity with $\min\{a, b\} \geq 3$ for which one of the primes is 3 are covered neither by Theorem 5 nor by that statement; we make no claim about whether they have been treated elsewhere. The earlier paper [6] of Jiang and Yang was revised on 30 October 2026, with a new title and an extension to several odd primes; according to its abstract and its Theorem A the primes with odd exponent are still required to be at least 5, and its Remark 4.2 still says that $q = 3$ is not covered, so it does not include the case of Theorem 2. All our references to the numbering of [6] are to its version v1. A formal proof in Lean of Roldán's conjecture for all distinct odd primes, including $q = 3$, was merged into the `trureturing` repository on 20 October 2026 [15]. It concerns only the exponents $(2, 3)$, that is, the case $r = s = 1$ of Theorems 1 and 2; we read its description, not its Lean code.

We searched again for this version on 7 November 2026, between 06:11 and 06:43 UTC: the arXiv (API, and the listings of math.CO, math.NT and math.SP up to those of 6 November 2026); Semantic Scholar, OpenAlex, Crossref, DataCite, Zenodo and zbMATH Open; MathDB; MathOverflow and Mathematics Stack Exchange, through the Stack Exchange API; GitHub commits, issues, pull requests, repositories and code; the file trees of 29 repositories of AI-assisted mathematics, and a text search of 28 clones of such repositories, dated up to 6 November 2026; and web search. Apart from the works above we found no other work on the maximal energy of these graphs since 25 September 2026, and in particular no treatment of the orders of equal parity with $\min\{a,b\} \geq 3$ in which one of the primes is 3. These paragraphs report what we read and the coverage of our searches, not a guarantee of priority.

Our proof replaces the semidefinite part of [6] by Theorem 12 below, an inequality for sign matrices that holds for all exponents and parities and all real $p \geq 5$, $q \geq 3$. By symmetry it also yields Theorem 1 without semidefinite certificates (Remark 13). Section 8 treats exponents of equal parity, and Section 9 proves Theorem 3 by applying the path identity and the potential of Sections 3 and 4 to the two rows of the sign matrix. Section 10 applies them to all rows of $T_a(p)Y$ at once, so that the states of the path are vectors built from the columns of the sign matrix Y. With this, Section 11 proves Theorem 4, and Section 12 gives a conditional reduction for larger exponents, of which we claim no instance. Section 13 describes the formal verification.

2 Weighted Ramanujan transforms

We use the notation of [6]. For real $x > 1$, an integer $k \geq 1$ and $0 \leq i, j \leq k$, let $T_k(x) = (t_{ij})$ be the symmetric matrix with

$$t_{ij} = \begin{cases} 0, & i+j \leq k-2, \\ -x^{k-1}(x-1), & i+j = k-1, \\ x^{2k-i-j-2}(x-1)^2, & i+j \geq k, \end{cases} \qquad (i,j < k),$$

$t_{ik} = t_{ki} = x^{k-i-1}(x-1)$ for $i < k$, and $t_{kk} = 1$. When x is prime, $t_{ij} = \varphi(x^{k-i})\, c_{x^{k-j}}(x^i)$, where $c_m(u)$ denotes Ramanujan's sum [6, (2.2)–(2.4)]. The row sums are $T_k(x)\mathbf{1} = x^k e_k$ [6, (2.5)], where e_k is the last standard basis vector. For a real matrix Z let $\|Z\|_1$ be the sum of the absolute values of its entries.

Proposition 6 ([6, Proposition 2.2])**.** *Let $p \neq q$ be primes, $a, b \geq 1$, $n = p^a q^b$, and let $X \in \{0,1\}^{(a+1)\times(b+1)}$ have $x_{ij} = 1$ exactly when $p^i q^j \in \mathcal{D}$. Then $\mathcal{E}(\mathrm{ICG}_n(\mathcal{D})) = \|T_a(p) X T_b(q)^{\mathsf{T}}\|_1$.*

Let $s_k = (1, -1, 1, \ldots, (-1)^k)^{\mathsf{T}}$ and define $d_k(x) = \|T_k(x)s_k\|_1$, and put $d_0(x) = 1$. By Remark 9 below, $d_k(x)$ agrees with the closed form in Theorem 1 for all real $x > 2$; see also [6, (3.6)].

3 A path identity

For $w \in \mathbb{R}^{k+1}$ define $z_{k-1} = w_k$ and, for $j = k-1, k-2, \ldots, 0$,

$$z_{j-1} = \frac{z_j + (x-1)w_j}{x}. \tag{1}$$

Thus z_j is a convex combination of $w_{j+1}, \ldots, w_k$, and each z_j is a linear function of w.

Lemma 7. *For $w \in \mathbb{R}^{k+1}$ we have*

$$\|T_k(x)w\|_1 = x^k\Big(|z_{-1}| + \sum_{j=0}^{k-1} |z_{j-1} - z_j|\Big).$$

Proof. Unwinding (1) gives $z_j = x^{-(k-j-1)} \sum_{i>j} \varphi(x^{k-i}) w_i$ with $\varphi(x^0) = 1$ and $\varphi(x^e) = x^{e-1}(x-1)$. The last row of $T_k(x)$ is $(\varphi(x^{k-i}))_i$, so $(T_k(x)w)_k = x^k z_{-1}$. For $u < k$ put $j = k-1-u$. Row u has entry $-x^{k-1}(x-1)$ in column j, entries $x^j(x-1)\varphi(x^{k-i})$ in columns $i > j$, and zeros elsewhere. Hence $(T_k(x)w)_u = x^{k-1}(x-1)(z_j - w_j)$, while $z_{j-1} - z_j = \frac{x-1}{x}(w_j - z_j)$. □

4 The one-dimensional potential

For $x > 2$ let $\mu_{-1} = 1$, $\mu_j = (x - 1 - \mu_{j-1})/x$ for $j \geq 0$, and $c_j = (x-1)(1+\mu_{j-1})/x$. By induction, $\mu_j \in [\frac{x-2}{x}, 1]$ for all j.

Lemma 8. *Let $x > 2$, let $y \in \{\pm 1\}^{k+1}$, and let z_j be defined by (1) with $w = y$. Then* $\operatorname{sgn} z_j = y_{j+1}$ *and* $\frac{x-2}{x} \leq |z_j| \leq 1$ *for* $-1 \leq j \leq k-1$, *and*

$$\frac{\|T_k(x)y\|_1}{x^k} = \sum_{j=0}^{k-1} c_j + \mu_{k-1} - \sum_{j \in \mathrm{NA}(y)} 2\mu_j |z_j|, \qquad \mathrm{NA}(y) = \{0 \leq j \leq k-1 : \ y_j = y_{j+1}\}.$$

Consequently $\|T_k(x)y\|_1 \leq d_k(x)$, with equality exactly when $y = \pm s_k$, and $d_k(x) = x^k(\sum_j c_j + \mu_{k-1})$. Moreover $(T_k(x)s_k)_k > 0$.

Proof. We have $z_{k-1} = y_k$, and since $x - 1 > 1 \geq |z_j|$, the sign of $z_{j-1} = (z_j + (x-1)y_j)/x$ is y_j and $|z_{j-1}| = (x - 1 \pm |z_j|)/x \in [\frac{x-2}{x}, 1]$. Put $f_j = |z_{j-1} - z_j| + \mu_{j-1}|z_{j-1}| - \mu_j|z_j|$. If $y_j = -\operatorname{sgn} z_j$, then $|z_{j-1} - z_j| = \frac{x-1}{x}(1 + |z_j|)$ and $|z_{j-1}| = \frac{x-1-|z_j|}{x}$; using $x\mu_j = x - 1 - \mu_{j-1}$ one finds $f_j = c_j$. If $y_j = \operatorname{sgn} z_j$, the same computation with $|z_{j-1} - z_j| = \frac{x-1}{x}(1 - |z_j|)$ and $|z_{j-1}| = \frac{x-1+|z_j|}{x}$ gives $f_j = c_j - 2\mu_j|z_j|$. Summing, the terms $\mu_j|z_j|$ telescope, and Lemma 7 gives the identity, since $\mu_{-1} = 1$ and $|z_{k-1}| = 1$. Every $j \in \mathrm{NA}(y)$ contributes a positive term $2\mu_j|z_j|$, and $\mathrm{NA}(y) = \emptyset$ exactly for $y = \pm s_k$. Finally $(T_k(x)s_k)_k = x^k z_{-1}$ and $\operatorname{sgn} z_{-1} = (s_k)_0 = 1$. □

Remark 9. Put $\delta_k(x) = x^k \mu_{k-1}$. Solving the recursion for μ gives $\delta_k(x) = (x^k(x-1) + 2(-1)^k)/(x+1) = x^k + 2\sum_{i<k}(-1)^{k-i}x^i$. For $y = s_k$, every step in the proof of Lemma 8 has $y_j = -\operatorname{sgn} z_j$, so $|z_{j-1}| = (x - 1 - |z_j|)/x$; starting from $|z_{k-1}| = 1 = \mu_{-1}$ this gives $|z_j| = \mu_{k-2-j}$, and in particular $(T_k(x)s_k)_k = x^k z_{-1} = \delta_k(x)$. Since the potential does not depend on k, Lemma 8 gives $d_{k+1}/x^{k+1} - d_k/x^k = c_k + \mu_k - \mu_{k-1} = 2\mu_k$, that is, $d_{k+1}(x) = x\, d_k(x) + 2\delta_{k+1}(x)$, with $d_0 = 1$. The closed form $(2k+1)x^k + 4\sum_{j=0}^{k-1}(-1)^{k-j}(j+1)x^j$ satisfies the same recursion and initial value, so it equals $d_k(x)$ for all $k \geq 0$.

5 A two-variable inequality

Lemma 10. *Let $P \geq \mu$ and $0 \leq \mu \leq 1$, and put $h(A,B) = P|A - B| + \mu|B + PA| - (P - \mu)|B| - P(1+\mu)|A|$ for real A, B. Then $h(A,B) \leq 2\mu \max\{0, |B| - P|A|\}$.*

Proof. As $h(-A,-B) = h(A,B)$ we may assume $A \geq 0$. If $0 \leq B \leq A$ then $h = -2(P-\mu)B \leq 0$. If $B > A$ then $h = 2(\mu B - PA) \leq 2\mu(B - PA)$ because $\mu \leq 1$. If $-PA \leq B < 0$ then $h = 0$. If $B < -PA$ then $h = 2\mu(|B| - PA)$. □

Lemma 11. *Let $q \geq 3$, $b \geq 1$, $N = T_b(q)$, $F(w) = \|Nw\|_1$ and $D = d_b(q)$. Let $P \geq 4$ and $\frac{3}{5} \leq \mu \leq 1$. For all $y \in \{\pm 1\}^{b+1}$ and $\zeta \in [-1,1]^{b+1}$,*

$$\Lambda(y,\zeta) := P\,F(y - \zeta) + \mu\,F(\zeta + Py) - (P - \mu)\,F(\zeta) \leq P(1+\mu)\,D,$$

with strict inequality unless $y = \pm s_b$.

Proof. Let $\xi_j(w)$ be the path (1) with $x = q$, $k = b$, and put $\tau_{-1}(w) = \xi_{-1}(w)$, $\tau_j(w) = \xi_{j-1}(w) - \xi_j(w)$ for $0 \le j < b$. These are linear, and $F(w) = q^b \sum_j |\tau_j(w)|$ by Lemma 7. With $A_j = \tau_j(y)$ and $B_j = \tau_j(\zeta)$,

$$\Lambda/q^b = P(1+\mu)F(y)/q^b + \textstyle\sum_j h(A_j, B_j).$$

By Lemma 8 (with $x = q$ and potential μ_j^q), it suffices to show $\sum_j h(A_j, B_j) \le \sum_{j \in \mathrm{NA}(y)} 2P(1+\mu)\mu_j^q x_j$, where $x_j = |\xi_j(y)| \in [\frac{q-2}{q}, 1]$, with strict inequality when $\mathrm{NA}(y) \neq \emptyset$. Since $\xi_j(\zeta)$ is a convex combination of entries of ζ, we have $|B_{-1}| \le 1$ and $|B_j| = \frac{q-1}{q}|\zeta_j - \xi_j(\zeta)| \le \frac{2(q-1)}{q}$. Also $|A_{-1}| \ge \frac{q-2}{q}$, and $|A_j| = \frac{q-1}{q}(1 + x_j)$ for $j \notin \mathrm{NA}(y)$, $|A_j| = \frac{q-1}{q}(1 - x_j)$ for $j \in \mathrm{NA}(y)$. We use Lemma 10, noting $P \ge 4 > \mu$.

- $j = -1$: $P|A_{-1}| \ge \frac{4(q-2)}{q} \ge \frac{4}{3} > |B_{-1}|$, so $h \le 0$.
- $j \ge 0$, $j \notin \mathrm{NA}(y)$: $P|A_j| \ge \frac{8(q-1)^2}{q^2} \ge \frac{2(q-1)}{q} \ge |B_j|$, so $h \le 0$.
- $j \in \mathrm{NA}(y)$: it suffices that $\mu|B_j| < P\beta(x_j)$, where $\beta(t) = \mu\frac{q-1}{q}(1-t) + (1+\mu)\mu_j^q t$. As β is affine it suffices to check the endpoints. For $q = 3$, $\beta(1) \ge \frac{8}{5}\cdot\frac{1}{3}$ and $\beta(\frac{1}{3}) \ge \frac{3}{5}\cdot\frac{4}{9} + \frac{8}{5}\cdot\frac{1}{9} = \frac{4}{9}$, so $P\beta \ge \frac{16}{9} > \frac{4}{3} \ge \mu|B_j|$. For general $q \ge 3$ the same comparison reduces to $P\beta(1) \ge \frac{32}{5}\cdot\frac{q-2}{q} > \frac{2(q-1)}{q}$ and $\frac{22}{5}q^2 - \frac{94}{5}q + \frac{104}{5} > 0$.

Summing over j proves the claim. □

6 The sign-matrix inequality

Theorem 12. *Let $p \ge 5$ and $q \ge 3$ be real, $a, b \ge 1$, $M = T_a(p)$ and $N = T_b(q)$. For every $Y \in \{\pm 1\}^{(a+1)\times(b+1)}$,*

$$\|MYN^{\mathsf{T}}\|_1 \le d_a(p)\, d_b(q),$$

with equality exactly when $Y = \pm s_a s_b^{\mathsf{T}}$.

Proof. Let $\bar{y}_0, \dots, \bar{y}_a$ be the rows of Y. Define $\zeta_{a-1} = \bar{y}_a$ and $\zeta_{k-1} = (\zeta_k + (p-1)\bar{y}_k)/p \in [-1,1]^{b+1}$. Column v of MYN^{T} is $Mw^{(v)}$ with $w_i^{(v)} = (N\bar{y}_i)_v$, and by linearity the path of $w^{(v)}$ is $((N\zeta_k)_v)_k$. Applying Lemma 7 with $x = p$ to every column and summing,

$$\|MYN^{\mathsf{T}}\|_1 = p^a\Big(F(\zeta_{-1}) + \sum_{k=0}^{a-1} F(\zeta_{k-1} - \zeta_k)\Big), \qquad F(w) = \|Nw\|_1.$$

Let μ_k be the potential of Lemma 8 for $x = p$ and put $f_k = F(\zeta_{k-1} - \zeta_k) + \mu_{k-1}F(\zeta_{k-1}) - \mu_k F(\zeta_k)$. Telescoping gives $\|MYN^{\mathsf{T}}\|_1/p^a = \sum_k f_k + \mu_{a-1}F(\bar{y}_a)$. Since $\zeta_{k-1} - \zeta_k = \frac{p-1}{p}(\bar{y}_k - \zeta_k)$ and $p\mu_k = p - 1 - \mu_{k-1}$, we get $pf_k = \Lambda(\bar{y}_k, \zeta_k)$ with $P = p - 1 \ge 4$ and $\mu = \mu_{k-1} \in [\frac{p-2}{p}, 1] \subseteq [\frac{3}{5}, 1]$. Lemma 11 gives $f_k \le c_k d_b(q)$, and Lemma 8 gives $F(\bar{y}_a) \le d_b(q)$; both are strict unless the row involved is $\pm s_b$, and $\mu_{a-1} > 0$. Hence $\|MYN^{\mathsf{T}}\|_1 \le p^a(\sum_k c_k + \mu_{a-1})d_b(q) = d_a(p)d_b(q)$. If equality holds, every row of Y is $\pm s_b$, so $Y = g s_b^{\mathsf{T}}$ with $g \in \{\pm 1\}^{a+1}$ and $\|MYN^{\mathsf{T}}\|_1 = \|Mg\|_1\, d_b(q)$; by Lemma 8 this equals $d_a(p)d_b(q)$ only for $g = \pm s_a$. □

Remark 13. Since $\|MYN^{\mathsf{T}}\|_1 = \|NY^{\mathsf{T}}M^{\mathsf{T}}\|_1$, the roles of the two factors may be exchanged, so Theorem 12 holds whenever $\min(p,q) \ge 3$ and $\max(p,q) \ge 5$. In particular it contains the sign-matrix inequality [6, Lemma 4.1] used for Theorem 1, and the deduction below then proves Theorem 1 as well. The semidefinite argument of [6] does not extend to $x = 3$ [6, Remark 4.2]; the proof above instead uses the integrality of the signs on both factors.

7 Proof of Theorem 2

Let $a = 2r$, $b = 2s+1$, $M = T_a(p)$, $N = T_b(3)$, let X be the divisor matrix of $\mathcal{D}$ (so $x_{ab} = 0$), and let $Y = 2X - J$ with J the all-ones matrix. By Proposition 6 and $M\mathbf{1} = p^a e_a$, $N\mathbf{1} = 3^b e_b$,

$$2\mathcal{E}(\mathrm{ICG}_n(\mathcal{D})) = \|MJN^{\mathsf{T}} + MYN^{\mathsf{T}}\|_1 = \|n\, e_a e_b^{\mathsf{T}} + MYN^{\mathsf{T}}\|_1 \le n + d_a(p)\, d_b(3),$$

by Theorem 12. For the checkerboard set, $Y^* = s_a s_b^{\mathsf{T}}$, $x^*_{00} = 1$, and $x^*_{ab} = 0$ because $a+b$ is odd, so $\mathcal{D}^*_{r,s}$ is an admissible nonempty set. The (a,b) entry of $MY^*N^{\mathsf{T}} = (Ms_a)(Ns_b)^{\mathsf{T}}$ is positive by Lemma 8, so it does not cancel against n, and $2\mathcal{E}(\mathrm{ICG}_n(\mathcal{D}^*_{r,s})) = n + \|Ms_a\|_1\|Ns_b\|_1 = n + d_a(p)\, d_b(3)$. If $\mathcal{E}(\mathrm{ICG}_n(\mathcal{D})) = \mathcal{E}(\mathrm{ICG}_n(\mathcal{D}^*_{r,s}))$, then $\|MYN^{\mathsf{T}}\|_1 = d_a(p)d_b(3)$, so $Y = \pm s_a s_b^{\mathsf{T}}$ by Theorem 12; since $y_{ab} = -1 = (s_a s_b^{\mathsf{T}})_{ab}$, we get $Y = Y^*$ and $\mathcal{D} = \mathcal{D}^*_{r,s}$. □

8 Exponents of equal parity

Let p, q be distinct odd primes, let $a, b \ge 1$ with $a+b$ even, $n = p^a q^b$, $M = T_a(p)$ and $N = T_b(q)$. For a set $\mathcal{D}$ of proper divisors of n let X and $Y = 2X - J$ be as in Section 7, so that $y_{ab} = -1$, and let $Z = MYN^{\mathsf{T}}$, with entries Z_{ij}. The last rows of M and N are nonnegative with sums p^a and q^b, so $|Z_{ab}| \le n$. Writing $t^- = \max\{-t, 0\}$, the computation of Section 7 gives

$$G(Y) := 2\mathcal{E}(\mathrm{ICG}_n(\mathcal{D})) - n = \sum_{(i,j)\ne(a,b)} |Z_{ij}| + Z_{ab} = \|Z\|_1 - 2(Z_{ab})^-. \tag{2}$$

Since $a+b$ is even, the sign matrix $s_a s_b^{\mathsf{T}}$ has entry $+1$ at (a,b) and violates the proper-divisor condition. The matrix $-s_a s_b^{\mathsf{T}}$, which corresponds to the set $\{p^i q^j : i+j \text{ odd}\}$, is admissible, and it attains the bound $\|Z\|_1 \le d_a(p)\, d_b(q)$ of Theorem 12. However, its entry $Z_{ab} = -(Ms_a)_a(Ns_b)_b = -\delta_a(p)\,\delta_b(q)$ is negative by Remark 9, and it cancels against n in the argument of Section 7. So the energy bound $\frac{1}{2}[n + d_a(p)\, d_b(q)]$ is not attained, and the maximiser changes. By (2), the set $\{p^i q^j : i+j \text{ odd}\}$ has energy $\frac{1}{2}[n + d_a(p)\, d_b(q)] - \delta_a(p)\,\delta_b(q)$.

Conjecture 14 (stated in earlier versions; proved here only for $\min\{a,b\} \le 2$)**.** *Let p, q be distinct odd primes, $a, b \ge 1$ with $a+b$ even, and $n = p^a q^b$. The maximal energy of an integral circulant graph of order n is*

$$\tfrac{1}{2}\,[n + d_a(p)\, d_b(q)] - \delta_a(p)\,\delta_b(q),$$

attained exactly by the sets $\{p^i q^j : i+j \text{ odd}\}$ and $\{p^i q^j : i+j \text{ even}\} \setminus \{n\}$.

By the computation above, the first of the two sets always attains the conjectured value. Since $d_1(x) = 3x - 4$, $\delta_1(x) = x - 2$, $d_2(x) = 5x^2 - 8x + 4$ and $\delta_2(x) = x^2 - 2x + 2$, Theorems 3 and 4 are the cases $\min\{a,b\} = 1$ and $\min\{a,b\} = 2$ of the conjecture, proved in Sections 9 and 11, and Theorem 5 is the case $\min\{a,b\} \le 2$. We prove the conjecture in no other case and keep it here as a historical statement. For $p, q \ge 5$ it is the statement of [7, Theorem A], whose proof we have not reviewed. For orders of equal parity with $\min\{a,b\} \ge 3$ in which one of the primes is 3 we make no claim (Section 1).

The number 15 is not used in this version. Proposition 15 of version 1.5 stated the conjecture, by computer-assisted certificates, for eleven exponent pairs. It is withdrawn. Its pairs with $\min\{a,b\} = 2$ are covered by Theorem 4; the pairs $(3,3)$, $(3,5)$, $(5,3)$ and $(4,4)$ are no longer claimed.

Remark 16. For $n = pq$, Theorem 3 gives the maximal energy $4(p-1)(q-1)$, attained exactly by $\{p, q\}$ and $\{1\}$. This can also be read off from known formulas. The graph $\mathrm{ICG}_{pq}(\{1\})$ is the unitary Cayley graph, whose energy $4\varphi(pq)$ was computed by Ilić [3], and $\mathcal{E}(\mathrm{ICG}_{pq}(\{p,q\})) = 4\varphi(pq)$ by [4, Theorem 4.2]. The graphs $\mathrm{ICG}_{pq}(\{p\})$ and $\mathrm{ICG}_{pq}(\{1,p\})$

are p disjoint copies of K_q and the complete q-partite graph with parts of size p (see also [4, Theorem 4.1]); both have energy $2p(q-1)$, and symmetrically for $\{q\}$ and $\{1,q\}$. Finally $\mathrm{ICG}_{pq}(\{1,p,q\})$ is the complete graph K_{pq}, of energy $2(pq-1)$. We have not found the maximal energy for $n=pq$ stated explicitly in the literature, in the searches described in Section 1. Le and Sander [5] study the maximum $\tilde{E}_{\max}(n)$ of the energy over *multiplicative* divisor sets; for their results for general n see [5, Theorem 2.1 and Corollary 2.1], which we neither restate nor formalise. We use this only for $n=p^aq^b$: a set $\mathcal{D}$ is multiplicative if $\mathcal{D}=\{p^iq^j:\ i\in A,\ j\in B\}$ for some $A\subseteq\{0,\dots,a\}$ and $B\subseteq\{0,\dots,b\}$, which for a nonempty set of proper divisors means that $\mathcal{D}$ is a product of a set of divisors of p^a and a set of divisors of q^b, and $\tilde{E}_{\max}(n)$ is the maximum over the multiplicative nonempty sets of proper divisors of n. These are admissible sets, so $\tilde{E}_{\max}(n)\le E_{\max}(n)$, the maximum over all of them. For $n=pq$ equality holds, but for $n=pq^3$ it fails: $E_{\max}(5\cdot3^3)=584>512=\tilde{E}_{\max}(5\cdot3^3)$ and $E_{\max}(3\cdot5^3)=1632>1472=\tilde{E}_{\max}(3\cdot5^3)$, and in both cases the maximisers are the non-multiplicative sets of Theorem 3. The values 512 and 1472 are proved in Lean; we give no written proof of them.

Remark 17. Other sets come close to the maximum. Let $(a,b)=(2,2)$, $q=3$ and $\mathcal{D}=\{1,3,p^2,3p^2,9p\}$, so that $Y=s_2y^{\mathsf{T}}$ with $y=(1,1,-1)^{\mathsf{T}}$ and $Z=(Ms_2)(Ny)^{\mathsf{T}}$. The path of y is $z_1=-1$, $z_0=\frac{1}{3}$, $z_{-1}=\frac{7}{9}$, and $\mathrm{NA}(y)=\{0\}$, so Lemma 8 gives $\|Ny\|_1=d_2(3)-2=23$, and $Z_{22}=9\,\delta_2(p)\,z_{-1}>0$. By (2), $G(Y)=23\,d_2(p)$, while the maximum of G is $25\,d_2(p)-10\,\delta_2(p)=G(Y)+4(p-3)$ (Theorem 25). So the energy of $\mathcal{D}$ is smaller than the maximal energy $62p^2-90p+40$ only by $2(p-3)$.

9 Proof of Theorem 3

Let p,q,m and $n=pq^m$ be as in Theorem 3; the case $n=p^mq$ follows by exchanging the names of the primes. Put $x=q$, $P=x-1$ and $R=p-1$. As p and q are distinct odd primes, either $x\ge5$ and $R\ge2$, or $x=3$ and $R\ge4$; only these inequalities are used. In both cases

$$R+1<P(R-1). \tag{3}$$

Let μ_j and c_j be the potential of Lemma 8 for x, and put $\rho=\mu_{m-1}$, $\hat{c}_j=c_j$ for $0\le j<m$, $\hat{c}_m=\rho$ and $\hat{D}=\sum_{j=0}^m\hat{c}_j$. By Lemma 8 and Remark 9, $\hat{D}=d_m(x)/x^m$ and $\rho=\delta_m(x)/x^m$. For $w\in\mathbb{R}^{m+1}$ let $z_j(w)$, $-1\le j\le m-1$, be the path (1) with $k=m$, and let $\nu(w)=\|T_m(x)w\|_1/x^m$. For a sign vector y put $\sigma(y)=\sum_{j\in\mathrm{NA}(y)}\mu_j|z_j(y)|$, so that $\nu(y)=\hat{D}-2\sigma(y)$ by Lemma 8. For a statement S, $[S]$ is 1 if S holds and 0 otherwise.

For $0\le\mu\le1$ let h_μ be the function h of Lemma 10 with $P=x-1$. Besides Lemma 10 we use three values, which follow from the definition and from the case analysis in the proof of Lemma 10: $h_\mu(0,B)=2\mu|B|$; $h_\mu(A,B)=-2(P-\mu)|B|$ if $AB>0$ and $|B|\le|A|$; and $h_\mu(A,B)=0$ if $AB\le0$ and $|B|\le P|A|$.

Let $\mu_\infty=P/(x+1)$. As $x\mu_\infty=P-\mu_\infty$, the recursion for μ gives $\mu_j-\mu_\infty=(-1/x)^{j+1}(1-\mu_\infty)$, where $1-\mu_\infty>0$. Hence, for $j\ge0$,

$$\mu_j>\mu_\infty \text{ if } j \text{ is odd},\qquad \mu_j<\mu_\infty \text{ if } j \text{ is even},\qquad \mu_0\le\mu_j\le\mu_1. \tag{4}$$

In particular $\rho<\mu_\infty$, because $m-1$ is even.

Lemma 18. *For every $w\in\mathbb{R}^{m+1}$,*

$$\nu(w)=\sum_{j=0}^{m}\hat{c}_j|w_j|+\frac{1}{x}\sum_{j=0}^{m-1}h_{\mu_{j-1}}(w_j,z_j(w)).$$

Proof. Write $z_j=z_j(w)$. By (1), $z_{j-1}-z_j=\frac{P}{x}(w_j-z_j)$ and $xz_{j-1}=z_j+Pw_j$. Together with $x\mu_j=P-\mu_{j-1}$ this gives, for $0\le j<m$,

$$|z_{j-1}-z_j|+\mu_{j-1}|z_{j-1}|-\mu_j|z_j|=\tfrac{1}{x}\big[h_{\mu_{j-1}}(w_j,z_j)+P(1+\mu_{j-1})|w_j|\big].$$

Summing over j, the potential terms telescope to $\mu_{-1}|z_{-1}| - \mu_{m-1}|z_{m-1}| = |z_{-1}| - \rho|w_m|$, and Lemma 7 gives the claim. □

Lemma 19. *For $y \in \{\pm1\}^{m+1}$ we have $\nu(y) - |z_{-1}(y)| \le \hat{D} - \rho$, with equality exactly when $y = \pm s_m$.*

Proof. By Lemma 7, the left side is $\sum_{j<m} |z_{j-1} - z_j|$, where $z_j = z_j(y)$. Let $\mu'_{-1} = 0$ and $\mu'_j = (P - \mu'_{j-1})/x$ for $j \ge 0$, so that $\mu'_j \in (0, P/x]$ for $j \ge 0$. The computation in the proof of Lemma 8 uses only this recursion of the potential, so it gives

$$|z_{j-1} - z_j| + \mu'_{j-1}|z_{j-1}| - \mu'_j|z_j| = c'_j - 2\mu'_j|z_j|\,[j \in \mathrm{NA}(y)], \qquad c'_j = P(1 + \mu'_{j-1})/x.$$

Summing over $j < m$ and using $\mu'_{-1} = 0$ and $|z_{m-1}| = 1$,

$$\sum_{j<m} |z_{j-1} - z_j| = \sum_{j<m} c'_j + \mu'_{m-1} - 2 \sum_{j \in \mathrm{NA}(y)} \mu'_j |z_j|.$$

All terms of the last sum are positive, so the left side is maximal exactly when $\mathrm{NA}(y) = \emptyset$, that is, when $y = \pm s_m$. For $y = s_m$ it equals $\nu(s_m) - |z_{-1}(s_m)| = \hat{D} - \rho$, by Lemma 8 and Remark 9. □

Lemma 20. *Let $y \in \{\pm1\}^{m+1}$ with $y_0 = y_m$. Then $(3R-1)\,\sigma(y) > (R-1)\rho$.*

Proof. As m is odd and $y_0 = y_m$, $y \ne \pm s_m$, so $\mathrm{NA}(y)$ is nonempty; let $J = \max \mathrm{NA}(y)$. Since y alternates on $J+1, \dots, m$, the recursion $|z_{j-1}| = (P - |z_j|)/x$ from the proof of Lemma 8 holds for $J < j \le m-1$, and starting from $|z_{m-1}| = 1 = \mu_{-1}$ it gives $|z_J(y)| = \mu_{m-2-J}$. Hence $\sigma(y) \ge \mu_J \mu_{m-2-J}$. If $J = m-1$, this is ρ, and $(3R-1)\rho > (R-1)\rho$. Otherwise J and $m-2-J$ are nonnegative and have opposite parity, since their sum is odd, so $\sigma(y) > \mu_0\mu_\infty$ by (4). As $(3R-1)(x-2) - (R-1)x = 2R(x-3) + 2 > 0$, we have $(3R-1)\mu_0 > R-1$, and therefore $(3R-1)\sigma(y) > (3R-1)\mu_0\mu_\infty \ge (R-1)\mu_\infty \ge (R-1)\rho$. □

Proof of Theorem 3. Let $\mathcal{D}$ be a set of proper divisors of n, and let u and v be the two rows of its sign matrix Y, so that $v_m = -1$. Since $T_1(p) = \left(\begin{smallmatrix} -R & R \\ R & 1 \end{smallmatrix}\right)$, the rows of $Z = T_1(p) Y T_m(x)^\mathsf{T}$ are $-R\,T_m(x)w^1$ and $T_m(x)w^2$, where $w^1 = u - v$ and $w^2 = Ru + v$. By the proof of Lemma 7, $Z_{1m} = x^m z_{-1}(w^2)$, so (2) gives $G(Y) = x^m \Phi$ with

$$\Phi = R\,\nu(w^1) + \nu(w^2) - 2(z_{-1}(w^2))^-.$$

As $d_1(p) = 3R-1$ and $\delta_1(p) = R-1$, the claimed maximal energy is $\frac{1}{2}(n + x^m\Theta)$ with $\Theta = (3R-1)\hat{D} - 2(R-1)\rho$. We show that $\Phi \le \Theta$, with equality exactly for $Y = s_1 s_m^\mathsf{T} - 2e_1 e_m^\mathsf{T}$ and $Y = -s_1 s_m^\mathsf{T}$. These are the sign matrices of the two sets in the theorem, which are nonempty sets of proper divisors of n because $1+m$ is even; so the theorem follows.

Let $\mathcal{A} = \{j : u_j = v_j\}$. For $j \in \mathcal{A}$ we have $w^1_j = 0$ and $|w^2_j| = R+1$; for $j \notin \mathcal{A}$ we have $|w^1_j| = 2$ and $|w^2_j| = R-1$. So $R|w^1_j| + |w^2_j|$ is $R+1$ on $\mathcal{A}$ and $3R-1$ off $\mathcal{A}$, and Lemma 18, applied to w^1 and w^2, gives

$$\Theta - \Phi = 2(R-1)\Big(\sum_{j \in \mathcal{A}} \hat{c}_j - \rho\Big) - H + 2(z_{-1}(w^2))^-, \tag{5}$$

where H is the sum of the $2m$ *cells* $\frac{R}{x}\, h_{\mu_{j-1}}(w^1_j, z_j(w^1))$ and $\frac{1}{x}\, h_{\mu_{j-1}}(w^2_j, z_j(w^2))$, $0 \le j < m$. Recall that $z_j(w)$ is a convex combination of $w_{j+1}, \dots, w_m$.

(a) For the w^2-cells, $|z_j(w^2)| \le R+1 < P(R-1) \le P|w^2_j|$ by (3), so they are ≤ 0 by Lemma 10.

(b) For the w^1-cells with $j \notin \mathcal{A}$, $|z_j(w^1)| \le 2 \le P|w^1_j|$, so they are ≤ 0.

(c) A w^1-cell with $j \in \mathcal{A}$ equals $\frac{R}{x} h_{\mu_{j-1}}(0, z_j(w^1)) = \frac{2R}{x} \mu_{j-1} |z_j(w^1)|$.

For $j \in \mathcal{A}$ with $j < m$ let

$$s_j = 2(R-1)c_j - \tfrac{2R}{x}\,\mu_{j-1}|z_j(w^1)| \;\geq\; \tfrac{2}{x}\big[(R-1)P(1+\mu) - 2R\mu\big], \qquad \mu = \mu_{j-1},$$

where we used $|z_j(w^1)| \leq 2$. The bracket is affine in μ, and it equals $(R-1)P > 0$ at $\mu = 0$ and $2[R(P-1) - P] > 0$ at $\mu = 1$; hence $s_j > 0$. Let $S = \sum_{j \in \mathcal{A},\, j<m} s_j$, let $C \geq 0$ be minus the sum of the cells in (a) and (b), and let $K = 2(z_{-1}(w^2))^- \geq 0$. Then (5) reads

$$\Theta - \Phi = S + C + K - 2(R-1)\rho\,[m \notin \mathcal{A}]. \tag{6}$$

Moreover $|z_0(w^2)| \leq R+1 < P|w_0^2|$, so $z_{-1}(w^2) = (z_0(w^2) + Pw_0^2)/x$ has the sign of $w_0^2 = Ru_0 + v_0$, which is the sign of u_0. Thus $K = 0$ if and only if $u_0 = 1$.

Case 1: $m \in \mathcal{A}$. Then $\Theta - \Phi = S + C + K \geq 0$. Suppose that equality holds. Then $S = 0$, so $\mathcal{A} = \{m\}$; $K = 0$, so $u_0 = 1$; and $C = 0$. Now $w_j^1 = 2u_j$ for $j < m$ and $w_m^1 = 0$. For $j \leq m-2$ we have $|Pw_{j+1}^1| = 2P > 2 \geq |z_{j+1}(w^1)|$, so $z_j(w^1) = (z_{j+1}(w^1) + Pw_{j+1}^1)/x$ has the sign of u_{j+1}, and $|z_j(w^1)| \leq 2 = |w_j^1|$. If $u_j = u_{j+1}$ for some $j \leq m-2$, the w^1-cell at j is therefore $-\frac{2R}{x}(P - \mu_{j-1})|z_j(w^1)| < 0$, contradicting $C = 0$. So u alternates on $0, \ldots, m-1$ and $u_0 = 1$; together with $u_m = v_m = -1$ and $v_j = -u_j$ for $j < m$, this means $Y = s_1 s_m^{\mathsf{T}} - 2e_1 e_m^{\mathsf{T}}$, as m is odd. Conversely, for this Y we have $\mathcal{A} = \{m\}$, $u = s_m$ and $S = K = 0$. The w^1-cells vanish: for $j \leq m-2$, $w_j^1 = 2u_j$ and $z_j(w^1)$ have opposite signs and $|z_j(w^1)| \leq P|w_j^1|$, and $z_{m-1}(w^1) = w_m^1 = 0$. The w^2-cells vanish too: w_i^2 has the sign of u_i for every i; by (3), $z_j(w^2) = (z_{j+1}(w^2) + Pw_{j+1}^2)/x$ has the sign of w_{j+1}^2 for $j \leq m-2$, and $z_{m-1}(w^2) = w_m^2$; since u alternates, w_j^2 and $z_j(w^2)$ have opposite signs, and $|z_j(w^2)| \leq P|w_j^2|$ by (a). Hence $\Phi = \Theta$.

Case 2: $m \notin \mathcal{A} \neq \emptyset$. Then $u_m = 1$ and $\mathcal{A} \subseteq \{0, \ldots, m-1\}$. By (6) it suffices to find $k \in \mathcal{A}$ with $s_k > 2(R-1)\rho$. Since $\rho < \mu_\infty = P/(P+2)$, the lower bound for s_k shows that this holds if

$$(R-1)P + \mu(P+2)\big[R(P-2) - P\big] \geq 0, \qquad \mu = \mu_{k-1}. \tag{7}$$

If $x \geq 5$, then $P \geq 4$ and $R(P-2) - P \geq P - 4 \geq 0$, so (7) holds for every $k \in \mathcal{A}$. If $x = 3$, then (7) reads $2(R-1) \geq 8\mu$. If some $k \in \mathcal{A}$ has $k \geq 1$, then $\mu_{k-1} \leq \mu_1 = \frac{5}{9}$ by (4), and $2(R-1) \geq 6 > \frac{40}{9}$. It remains to treat $x = 3$ and $\mathcal{A} = \{0\}$. Here $\mu_\infty = \frac{1}{2}$ and $c_0 = \frac{4}{3}$, and since $z_0(w)$ depends only on $w_1, \ldots, w_m$, we have $z_0(w^1) = 2z_0(u)$ and $s_0 = \frac{2}{3}\big[4(R-1) - 2R|z_0(u)|\big]$. If $m = 1$, then $|z_0(u)| = 1$, $\rho = \mu_0 = \frac{1}{3}$ and $s_0 = \frac{2}{3}(2R-4) > \frac{2}{3}(R-1) = 2(R-1)\rho$. Let $m \geq 3$. If $|z_0(u)| \leq \frac{5(R-1)}{4R}$, then $s_0 \geq R - 1 > 2(R-1)\rho$. Otherwise $|z_0(u)| > \frac{15}{16}$. If $u_1 \neq u_2$, then $u_1 = -\operatorname{sgn} z_1(u)$ by Lemma 8, and $|z_0(u)| = (2 - |z_1(u)|)/3 \leq \frac{5}{9}$; hence $u_1 = u_2$. Then $1 \notin \mathcal{A}$, and the arguments $w_1^1 = 2u_1$ and $z_1(w^1) = 2z_1(u)$ of the w^1-cell at $j = 1$ have the same sign, with $|z_1(w^1)| \leq |w_1^1|$. This cell equals $-\frac{2R}{3}(P - \mu_0)|z_1(w^1)| = -\frac{20R}{9}|z_1(u)| \leq -\frac{20R}{27}$, so $C \geq \frac{20R}{27}$. With $s_0 \geq \frac{2}{3}(2R-4)$ and $2(R-1)\rho < R - 1$,

$$\Theta - \Phi \geq \tfrac{2}{3}(2R-4) + \tfrac{20R}{27} - 2(R-1)\rho > \tfrac{56R-72}{27} - (R-1) = \tfrac{29R-45}{27} > 0.$$

Case 3: $\mathcal{A} = \emptyset$, that is, $v = -u$. Then $u_m = 1$, $w^1 = 2u$, $w^2 = (R-1)u$ and $\Phi = (3R-1)\nu(u) - 2(R-1)(z_{-1}(u))^-$. If $u_0 = 1$, then $z_{-1}(u) > 0$ by Lemma 8, and $\Phi = (3R-1)(\hat{D} - 2\sigma(u)) < \Theta$ by Lemma 20, since $u_0 = u_m$. If $u_0 = -1$, then $(z_{-1}(u))^- = |z_{-1}(u)|$ and

$$\Phi = (R+1)\,\nu(u) + 2(R-1)\big(\nu(u) - |z_{-1}(u)|\big) \leq (R+1)\hat{D} + 2(R-1)(\hat{D} - \rho) = \Theta$$

by Lemmas 8 and 19, with equality exactly when $u = \pm s_m$, that is, $u = -s_m$. This is $Y = -s_1 s_m^{\mathsf{T}}$.

The three cases cover all Y, which proves the theorem. For $m = 1$ the value is $4(p-1)(q-1)$, since $d_1(q) = 3q - 4$ and $\delta_1(q) = q - 2$. □

10 The column path

In this section and the next two, the exponent $a \geq 2$ of p is fixed, $b \geq 1$ is any integer with $a+b$ even, and p, q are real numbers with

$$p \geq 5 \text{ and } q \geq 3, \qquad \text{or} \qquad p = 3 \text{ and } q \geq 5. \tag{8}$$

Every ordered pair of distinct odd primes satisfies (8). We use the notation of Section 9 with m replaced by b: $x = q$, $P = x - 1$ and $R = p - 1$; μ_j and c_j form the potential of Lemma 8 for x, and $\mu_\infty = P/(x+1)$; $\rho = \mu_{b-1}$, $\hat{c}_j = c_j$ for $j < b$, $\hat{c}_b = \rho$ and $\hat{D} = \sum_{j=0}^{b} \hat{c}_j$, so that $\hat{D} = d_b(q)/x^b$ and $\rho = \delta_b(q)/x^b$; for $w \in \mathbb{R}^{b+1}$, $z_j(w)$ is the path (1) with $k = b$ and $\nu(w) = \|T_b(x)w\|_1/x^b$; and h_μ is the function h of Lemma 10 with $P = x - 1$. Lemma 18 and (4) hold with the same proofs, which do not use that m is odd. By Lemma 8 and (4),

$$\mu_0 \leq \mu_j \leq 1 \quad (j \geq -1), \qquad \mu_j \leq \mu_1 \quad (j \geq 0), \qquad \mu_0 = \frac{P-1}{x}, \quad \mu_1 = \frac{P^2+1}{x^2}. \tag{9}$$

By (8), either $R \geq 4$ and $P \geq 2$, or $R = 2$ and $P \geq 4$; in particular $P \geq 2$ and $PR \geq 8$.

Values of the cell function. Besides the three values of h_μ listed in Section 9, the case analysis in the proof of Lemma 10, together with $h_\mu(-A, -B) = h_\mu(A, B)$, gives

$$h_\mu(A,B) = \begin{cases} 2(\mu|B| - P|A|) & \text{if } AB > 0 \text{ and } |B| \geq |A|, \\ 2\mu(|B| - P|A|) & \text{if } AB \leq 0 \text{ and } |B| \geq P|A|. \end{cases} \tag{10}$$

Consequently, for $0 \leq \mu \leq 1$,

$$h_\mu(A,B) < 0 \qquad \text{if } AB > 0 \text{ and } |B| < P|A|, \tag{11}$$

since the value is then $-2(P-\mu)|B|$ or $2(\mu|B| - P|A|) \leq 2(|B| - P|A|)$. The assumption $|B| < P|A|$ is needed here: $h_\mu(A, PA/\mu) = 0$ for $A > 0$ and $0 < \mu \leq 1$.

The matrix M. Let $M = T_a(p)$, $\alpha = Ms_a$ and $u = Me_a$, and let ω_i be the sum of the absolute values of the entries of row i of M. Thus $\|\alpha\|_1 = d_a(p)$, $\alpha_a = \delta_a(p)$ by Remark 9, and $|(M\zeta)_i| \leq \omega_i$ for $\zeta \in [-1,1]^{a+1}$. Let $\mu_j^{(p)}$ denote the potential of Lemma 8 for p in place of x. For $i < a$,

$$u_i = p^{a-1-i}R, \qquad \omega_i = 2p^{a-1}R, \qquad \alpha_i = (-1)^{a-i}p^{a-1}R(1 + \mu_{i-1}^{(p)}), \tag{12}$$

and $u_a = 1$, $\omega_a = p^a$. Indeed, by the proof of Lemma 7 with $x = p$ and $k = a$, row $i < a$ of M has the entry $-p^{a-1}R$ in column $l = a - 1 - i$ and nonnegative entries with sum $p^{a-1}R$ in the columns after l, and $\alpha_i = p^{a-1}R\,(z_l - (s_a)_l)$ for the path z of s_a, where $\operatorname{sgn} z_l = (s_a)_{l+1}$ and $|z_l| = \mu_{a-2-l}^{(p)}$ by Lemma 8 and Remark 9; the last row of M is $(p^{a-1}R, \ldots, pR, R, 1)$. Moreover $\delta_a(p) = p^a\mu_{a-1}^{(p)} \geq p^{a-1}(p-2) \geq 3$. Hence

$$2u_i \leq |\alpha_i| \quad (i < a), \qquad \delta_a(p) \geq 3, \qquad P\,\delta_a(p) > p^a, \qquad \omega_i \leq P|\alpha_i| \quad (0 \leq i \leq a). \tag{13}$$

The first inequality reads $2 \leq p^i(1+\mu_{i-1}^{(p)})$, with equality for $i = 0$. Next, $P\delta_a(p) \geq 2p^{a-1}(p-2) > p^a$ if $p \geq 5$, and $P\delta_a(p) \geq 4 \cdot 3^{a-1} > 3^a$ if $p = 3$. Finally $\omega_i \leq 2|\alpha_i| \leq P|\alpha_i|$ for $i < a$, and $\omega_a = p^a < P\delta_a(p)$.

Columns. For $c \in \{\pm 1\}^{a+1}$ put $\Delta(c) = d_a(p) - \|Mc\|_1$. By Lemma 8, $\Delta(c) \geq 0$, with equality exactly for $c = \pm s_a$. Let $Y \in \{\pm 1\}^{(a+1)\times(b+1)}$ have columns $c_0, \dots, c_b$. Define $\zeta_{b-1} = c_b$ and $\zeta_{j-1} = (\zeta_j + Pc_j)/x$ for $j = b-1, \dots, 0$; this is the path (1) applied to the sequence of columns. Each ζ_j is a convex combination of $c_{j+1}, \dots, c_b$, so $\zeta_j \in [-1,1]^{a+1}$, and if $r_0, \dots, r_a$ are the rows of MY, then $z_j(r_i) = (M\zeta_j)_i$ by linearity. For $0 \leq \mu \leq 1$, $c \in \{\pm 1\}^{a+1}$ and $\zeta \in [-1,1]^{a+1}$ let

$$\psi_\mu(c,\zeta) = P(1+\mu)\,\Delta(c) - \sum_{i=0}^{a} h_\mu((Mc)_i, (M\zeta)_i);$$

we call the terms $h_\mu((Mc)_i, (M\zeta)_i)$ the *cells* of $\psi_\mu(c,\zeta)$. As $h_\mu(-A,-B) = h_\mu(A,B)$, we have $\psi_\mu(-c,-\zeta) = \psi_\mu(c,\zeta)$. For $0 \leq j < b$ put $\psi_j = \psi_{\mu_{j-1}}(c_j, \zeta_j)$.

Let $N = T_b(q)$ and $\Theta = d_a(p)\,d_b(q) - 2\delta_a(p)\,\delta_b(q)$, and let $G(Y)$ be the middle expression in (2) with $Z = MYN^{\mathsf{T}}$. Let $Y^- = -s_a s_b^{\mathsf{T}}$ and $Y^+ = s_a s_b^{\mathsf{T}} - 2e_a e_b^{\mathsf{T}}$. The columns of Y^- are $(-1)^{j+1}s_a$; those of Y^+ are $(-1)^j s_a$ for $j < b$ and $g_+ = (-1)^a s_a - 2e_a$ for $j = b$, as $(-1)^b = (-1)^a$. By Section 8, $G(Y^-) = \Theta$.

Lemma 21. *For every $Y \in \{\pm 1\}^{(a+1)\times(b+1)}$,*

$$x^{1-b}(\Theta - G(Y)) = \sum_{j=0}^{b-1} \psi_j + x\rho(\Delta(c_b) - 2\delta_a(p)) + 2x\big((M\zeta_{-1})_a\big)^-.$$

Proof. Row i of Z is $(Nr_i)^{\mathsf{T}}$, so the absolute values of its entries have sum $x^b\nu(r_i)$, and $Z_{ab} = (Nr_a)_b = x^b z_{-1}(r_a) = x^b(M\zeta_{-1})_a$ by the proof of Lemma 7. Hence $G(Y) = x^b[\sum_i \nu(r_i) - 2((M\zeta_{-1})_a)^-]$ by (2). We apply Lemma 18 to each r_i and add. As $\sum_i |(r_i)_j| = \|Mc_j\|_1 = d_a(p) - \Delta(c_j)$, $z_j(r_i) = (M\zeta_j)_i$ and $\hat{c}_j = P(1+\mu_{j-1})/x$ for $j < b$, this gives

$$x^{-b}G(Y) = d_a(p)\hat{D} - \sum_{j=0}^{b} \hat{c}_j\Delta(c_j) + \frac{1}{x}\sum_{j=0}^{b-1}\sum_{i=0}^{a} h_{\mu_{j-1}}((Mc_j)_i, (M\zeta_j)_i) - 2\big((M\zeta_{-1})_a\big)^-.$$

Since $x^{-b}\Theta = d_a(p)\hat{D} - 2\delta_a(p)\rho$ and $\hat{c}_b = \rho$, subtracting and multiplying by x proves the lemma. □

Lemma 22. *Let $0 \leq \mu \leq 1$, $c \in \{\pm 1\}^{a+1}$ and $\zeta \in [-1,1]^{a+1}$. Then*

$$\psi_\mu(c,\zeta) \geq P(1+\mu)\Delta(c) - 2\mu\sum_{i=0}^{a}\max\{0,\ |(M\zeta)_i| - P|(Mc)_i|\}.$$

If $c = \pm s_a$, all cells of $\psi_\mu(c,\zeta)$ are ≤ 0; so $\psi_\mu(c,\zeta) \geq 0$, with equality exactly when all cells vanish.

Proof. The bound follows from Lemma 10, which applies as $P \geq 2 > \mu$. If $c = \pm s_a$, then $\Delta(c) = 0$ and $|(M\zeta)_i| \leq \omega_i \leq P|\alpha_i| = P|(Mc)_i|$ by (13), so every cell is ≤ 0 by Lemma 10. □

Lemma 23. *Let $g \in \{\pm 1\}^{a+1}$ and $0 \leq k \leq b$, and suppose that $c_b = g$ and $c_{b-i} = (-1)^i g$ for $1 \leq i \leq k$. Then $\zeta_{b-1-i} = (-1)^i\mu_{i-1}g$ for $0 \leq i \leq k$, and $\psi_{b-i} = P(1+\mu_{b-i-1})\Delta(g)$ for $1 \leq i \leq k$. In particular, if $c_l = (-1)^{l+1}s_a$ for $J < l \leq b$, then $\zeta_J = (-1)^J\mu_{b-2-J}s_a$.*

Proof. The formula for ζ_{b-1-i} holds for $i = 0$ as $\mu_{-1} = 1$, and if it holds for $i-1$, then $\zeta_{b-1-i} = ((-1)^{i-1}\mu_{i-2}g + P(-1)^i g)/x = (-1)^i\mu_{i-1}g$. For $1 \leq i \leq k$, the cells of ψ_{b-i} have the arguments $A = (-1)^i Mg$ and $B = M\zeta_{b-i} = -\mu_{i-2}A$, so that $A_l B_l \leq 0$ and $|B_l| \leq |A_l| \leq P|A_l|$ for every l, and all cells vanish. For the last claim take $g = (-1)^{b+1}s_a$ and $i = b-1-J$. □

Lemma 24. *We have $\Delta(g_+) = 2\delta_a(p)$. For $Y = Y^+$, every ψ_j vanishes and $(M\zeta_{-1})_a > 0$, so that $G(Y^+) = \Theta$. Conversely, if $c_b = g_+$ and, for every $j < b$, $c_j = \pm s_a$ and $\psi_j = 0$, then $Y = Y^+$.*

Proof. By (12), $(Mg_+)_i = (-1)^a\alpha_i - 2u_i = (-1)^i(|\alpha_i| - 2(-1)^i u_i)$, and $|\alpha_i| - 2(-1)^i u_i \geq 0$ by (13). Hence $\|Mg_+\|_1 = \sum_i |\alpha_i| - 2\sum_i (-1)^i u_i = d_a(p) - 2s_a^{\mathsf{T}} M e_a = d_a(p) - 2\delta_a(p)$, as M is symmetric.

Let $Y = Y^+$. By induction on j downwards, $\zeta_j = (-1)^{j+1}\mu_{b-2-j}s_a - 2x^{j+1-b}e_a$ for $-1 \leq j \leq b-1$. At a column $j < b$, the cells of ψ_j have the arguments $A_i = (-1)^j\alpha_i$ and $B_i = (M\zeta_j)_i$, and with $\sigma_i = \operatorname{sgn} A_i = (-1)^{j+a-i}$,

$$\sigma_i B_i = -\mu_{b-2-j}|\alpha_i| - 2(-1)^{j+a-i}x^{j+1-b}u_i.$$

We claim that $-P|A_i| \leq \sigma_i B_i \leq 0$, so that every cell vanishes. The upper bound holds if $\mu_{b-2-j}|\alpha_i| \geq 2x^{j+1-b}u_i$: for $j = b-1$ this is $|\alpha_i| \geq 2u_i$, and for $j \leq b-2$ it follows from $\mu_{b-2-j} \geq \mu_0 = (P-1)/x$, $x^{j+1-b} \leq 1/x$ and $(P-1)|\alpha_i| \geq |\alpha_i| \geq 2u_i$, where $|\alpha_a| = \delta_a(p)$ and $u_a = 1$. The lower bound holds as $\mu_{b-2-j}|\alpha_i| + 2u_i \leq 2|\alpha_i| \leq P|\alpha_i|$. Since $\Delta(\pm s_a) = 0$, every ψ_j vanishes. Moreover $(M\zeta_{-1})_a = \mu_{b-1}\delta_a(p) - 2x^{-b} \geq ((P-1)\delta_a(p) - 2)/x > 0$. By Lemma 21 and $\Delta(g_+) = 2\delta_a(p)$, $G(Y^+) = \Theta$.

Conversely, let $c_b = g_+$, and let $c_j = \varepsilon_j s_a$ with $\varepsilon_j = \pm 1$ and $\psi_j = 0$ for $j < b$; by Lemma 22, all cells of every ψ_j vanish. Put $\beta_j = (M\zeta_j)_a$, so that $|\beta_j| \leq \omega_a = p^a$. We show $\varepsilon_j = (-1)^j$ by induction on j downwards. The cell of ψ_j at $i = a$ has the arguments $A = \varepsilon_j\delta_a(p)$ and $B = \beta_j$, and by (11) it is negative if $AB > 0$ and $|B| < P|A|$. So it suffices to check these two conditions for $\varepsilon_j = (-1)^{j+1}$. For $j = b-1$, $\beta_{b-1} = (Mg_+)_a = (-1)^b\delta_a(p) - 2$ has the sign $(-1)^b$, and $|\beta_{b-1}| \leq \delta_a(p) + 2 < P\delta_a(p)$. For $j \leq b-2$, if $\varepsilon_{j+1} = (-1)^{j+1}$, then $\beta_j = (\beta_{j+1} + (-1)^{j+1}P\delta_a(p))/x$ has the sign $(-1)^{j+1}$, since $|\beta_{j+1}| \leq p^a < P\delta_a(p)$, and $|\beta_j| \leq p^a < P\delta_a(p)$. Hence $\varepsilon_j = (-1)^j$ for all $j < b$, that is, $Y = Y^+$. □

11 Proof of Theorem 4

Theorem 25. *Let $b \geq 2$ be even, let p, q be real numbers with $p \geq 5$ and $q \geq 3$, or with $p = 3$ and $q \geq 5$, and put $\Theta = d_2(p)\,d_b(q) - 2\delta_2(p)\,\delta_b(q)$. Every $Y \in \{\pm 1\}^{3\times(b+1)}$ with $y_{2b} = -1$ satisfies $G(Y) \leq \Theta$, where $G(Y)$ is the middle expression in (2) with $Z = T_2(p)\,Y\,T_b(q)^{\mathsf{T}}$, and equality holds exactly for $Y = -s_2 s_b^{\mathsf{T}}$ and $Y = s_2 s_b^{\mathsf{T}} - 2e_2 e_b^{\mathsf{T}}$.*

Proof of Theorem 4. Let $n = p^2q^b$. By (2), $2\mathcal{E}(\mathrm{ICG}_n(\mathcal{D})) = n + G(Y)$ for the sign matrix Y of a set $\mathcal{D}$ of proper divisors of n, and $y_{2b} = -1$. The two matrices in Theorem 25 are the sign matrices of $\{p^iq^j : i+j \text{ odd}\}$ and of $\{p^iq^j : i+j \text{ even}\} \setminus \{n\}$. These are nonempty sets of proper divisors of n: the first contains p, and it does not contain n because $2+b$ is even, and the second contains 1. Distinct odd primes satisfy the hypothesis of Theorem 25, so the maximal energy is $\frac{1}{2}(n+\Theta)$, attained exactly by these two sets. By Remark 9, $d_2(p) = 5p^2 - 8p + 4$ and $\delta_2(p) = p^2 - 2p + 2$. The case $n = p^bq^2$ follows by exchanging the names of the primes. □

We now prove Theorem 25, with the notation of Section 10 for $a = 2$. Here

$$M = T_2(p) = \begin{pmatrix} 0 & -pR & pR \\ -pR & R^2 & R \\ pR & R & 1 \end{pmatrix},$$

and we put $g_1 = s_2 = (1,-1,1)^{\mathsf{T}}$, $g_2 = (1,1,-1)^{\mathsf{T}}$, $g_3 = (1,-1,-1)^{\mathsf{T}} = g_+$ and $g_4 = (1,1,1)^{\mathsf{T}}$. Using $p = R+1$,

$$Mg_1 = \begin{pmatrix} 2pR \\ -2R^2 \\ R^2+1 \end{pmatrix}, \quad Mg_2 = \begin{pmatrix} -2pR \\ -2R \\ R^2+2R-1 \end{pmatrix}, \quad Mg_3 = \begin{pmatrix} 0 \\ -2pR \\ R^2-1 \end{pmatrix}, \quad Mg_4 = \begin{pmatrix} 0 \\ 0 \\ p^2 \end{pmatrix}.$$

Every $c \in \{\pm 1\}^3$ equals $\pm g_k$ for exactly one k. As $d_2(p) = 5R^2 + 2R + 1$ and $\delta_2(p) = R^2 + 1$, we get $\Delta(\pm g_k) = \Delta_k$ with

$$\Delta_1 = 0, \qquad \Delta_2 = 2(R-1)^2, \qquad \Delta_3 = 2(R^2+1) = 2\delta_2(p), \qquad \Delta_4 = 4R^2. \tag{14}$$

Moreover $\omega = (2pR, 2pR, p^2)$, and

$$p^2 < P(R^2 + 2R - 1), \qquad p^2 < P(R^2 - 1), \tag{15}$$

because $2(R^2 + 2R - 1) - p^2 = (R+3)(R-1) > 0$, $2(R^2-1) - p^2 = (R+1)(R-3) > 0$ if $R \geq 4$, and $P(R^2-1) \geq 12 > 9 = p^2$ if $R = 2$.

Lemma 26. *Let $0 \leq \mu \leq 1$ and $\zeta \in [-1,1]^3$, and put $B = M\zeta$. If $c \neq \pm s_2$, then $\psi_\mu(c,\zeta) > 0$. Moreover*

$$\psi_\mu(\pm g_3, \zeta) \geq 2P(1+\mu)(R^2+1) - 2\mu|B_0|, \qquad \psi_\mu(\pm g_4, \zeta) \geq 4PR^2(1+\mu) - 2\mu(|B_0| + |B_1|). \tag{16}$$

Proof. We use Lemma 22, $|B_i| \leq \omega_i$ and (15). For $c = \pm g_2$ we have $|Mc| = (2pR, 2R, R^2+2R-1)$, so the terms with $i = 0, 2$ of the sum in Lemma 22 vanish, and $\psi_\mu(c,\zeta) \geq 2P(1+\mu)(R-1)^2 - 4\mu R \max\{0, p - P\}$. If $P \geq p$, this is positive. Otherwise $R \geq 4$ by (8), and the bound is affine in μ, positive at $\mu = 0$, and at $\mu = 1$ equal to $4[P(R^2 - R + 1) - R(R+1)] \geq 4(R-1)(R-2) > 0$. For $c = \pm g_3$ we have $|Mc| = (0, 2pR, R^2 - 1)$, so the terms with $i = 1, 2$ vanish, which gives the first bound in (16); with $|B_0| \leq 2pR$ and $P \geq 2$ its right side is at least $4(1+\mu)(R^2+1) - 4\mu pR = 4(R^2+1) - 4\mu(R-1) > 0$. For $c = \pm g_4$ we have $|Mc| = (0, 0, p^2)$, which gives the second bound; its right side is at least $8R^2(1+\mu) - 8\mu pR = 8R(R-\mu) > 0$. □

Lemma 27. *Let $R \geq 4$ and $P \geq 2$, or $R = 2$ and $P \geq 4$, and put $T = (R^2+1)(P^2+1)$. Then:*

(i) $(P-1)^2(5R^2 + 2R + 1) > T$;

(ii) $P(P+1)(R-1)^2 + (P-1)^2(3R^2+1) > T$;

(iii) $(R^2+1)(P-1) + (P+1)\big[2P(R^2 - R + 1) - 3R^2 - 1\big] > 0$;

(iv) $P(P+1)(R-1)^2 + 2(P-1)^2R(R+1) > T$;

(v) $(P+1)\big[P(R-1)^2 + 2P^2(R+1)\big] > T$;

(vi) $(P+1)\big[2PR(R-3) + 2P^2(R+1)\big] > T$ *if* $R \geq 4$;

(vii) $P(P+1)^2(R^2+1) > 2R(R+1)(P^2+1)$;

(viii) $2R(PR-1)(P+1) > T$.

Proof. The function $f(P) = (P-1)^2/(P^2+1) = 1 - 2P/(P^2+1)$ increases for $P \geq 1$, and $f(2) = \frac{1}{5}$, $f(4) = \frac{9}{17}$. (i) We have $5R^2 + 2R + 1 - 5(R^2+1) = 2(R-2) \geq 0$ and $5(P-1)^2 - (P^2+1) = 2(2P-1)(P-2) \geq 0$, so $(P-1)^2(5R^2+2R+1) \geq 5(P-1)^2(R^2+1) \geq T$, and one of the two inequalities is strict. (ii) The left side minus T equals $(R-1)^2(P-1) + (P-1)^2(3R^2+1) - 2R(P^2+1)$, and $(P-1)^2(3R^2+1) = f(P)(3R^2+1)(P^2+1) \geq 2R(P^2+1)$: if $R \geq 4$, then $f(P) \geq \frac{1}{5}$ and $3R^2 + 1 - 10R = R(3R-10) + 1 > 0$; if $R = 2$, then $f(P) \geq \frac{9}{17}$ and $\frac{9}{17} \cdot 13 > 4$. (iii) If $R \geq 4$, then $2P(R^2 - R + 1) - 3R^2 - 1 \geq 4(R^2 - R + 1) - 3R^2 - 1 = (R-1)(R-3) > 0$; if $R = 2$, it equals $6P - 13 > 0$. (iv) The left side minus T equals $(R-1)^2(P-1) + 2R\big[(P-1)^2(R+1) - (P^2+1)\big]$, and $f(P)(R+1) \geq 1$ both for $R \geq 4$ and for $R = 2$. (v) The left side minus T equals $(R-1)^2(P-1) + 2\big[R(P^3-1) + P^3 + P^2\big] > 0$. (vi) As $P(P+1) \geq P^2 + 1$, $R \geq 3$, $P^2 \geq \frac{4}{5}(P^2+1)$ and $P + 1 \geq 3$, the left side is at least $(P^2+1)\big[2R(R-3) + \frac{24}{5}(R+1)\big]$, and $2R(R-3) + \frac{24}{5}(R+1) - (R^2+1) = R^2 - \frac{6}{5}R + \frac{19}{5} > 0$. (vii) $P(P+1)^2 = (P+2)(P^2+1) - 2$, so $P(P+1)^2 \geq \frac{18}{5}(P^2+1)$ for $P \geq 2$, and $\frac{18}{5}(R^2+1) - 2R(R+1) = \frac{1}{5}(8R^2 - 10R + 18) > 0$. (viii) Twice the left side minus $2T$ equals $PR(PR-4) + P^2(R^2-2) + (4PR^2 - 2R^2 - 4R - 2)$, where $PR \geq 8$, $R \geq 2$ and $4PR^2 - 2R^2 - 4R - 2 \geq 2(3R+1)(R-1) > 0$. □

Lemma 28. *Let b be even, $0 \le J < b$, $\mu = \mu_{J-1}$ and $m = \mu_{b-2-J}$. For every $c \in \{\pm 1\}^3$ other than $-s_2$, $\psi_\mu(c, ms_2) > 2x\delta_2(p)\rho$.*

Proof. By (9), $\mu, m \in [\mu_0, 1]$ and $\rho \le \mu_1 = (P^2+1)/x^2$. Put $T = (R^2+1)(P^2+1)$ and $\psi = \psi_\mu(c, ms_2)$. Then $2x\delta_2(p)\rho \le 2T/x$, so it suffices to show $x\psi > 2T$; we do so except in one subcase of (d). We use

$$x(P-\mu)m \ge (P-1)^2, \tag{17}$$

which holds as $P - \mu \ge P - 1 \ge 0$ and $xm \ge x\mu_0 = P - 1$. Let $A = Mc$ and $B = M(ms_2) = m\,(2pR, -2R^2, R^2+1)$, and write $h_i = h_\mu(A_i, B_i)$; the cells are computed with the values of h_μ in Section 9 and (10), using $0 \le m \le 1$.

(a) $c = s_2$. For each i, $A_iB_i \ge 0$ and $|B_i| \le |A_i|$, so $h_i = -2(P-\mu)m|A_i|$ and $\psi = 2(P-\mu)m\,d_2(p)$. By (17) and Lemma 27(i), $x\psi \ge 2(P-1)^2(5R^2+2R+1) > 2T$.

(b) $c = g_2$, so $A = (-2pR, -2R, R^2+2R-1)$. Here $h_0 = 0$, as $A_0B_0 \le 0$ and $|B_0| \le P|A_0|$, and $h_2 = -2(P-\mu)m(R^2+1)$, as $A_2B_2 \ge 0$ and $|B_2| \le |A_2|$; also $A_1B_1 \ge 0$. If $mR \le 1$, then $|B_1| \le |A_1|$, $h_1 = -4(P-\mu)mR^2$ and $\psi = 2P(1+\mu)(R-1)^2 + 2(P-\mu)m(3R^2+1)$, so $x\psi > 2T$ by $\mu \ge 0$, (17) and Lemma 27(ii). If $mR > 1$, then $h_1 = 4R(\mu mR - P)$ by (10), and expanding gives

$$x\psi - 2T = 2(R^2+1)(P-1) + 2Px(R-1)^2\mu + 2Px(R^2+1)m - 2x(3R^2+1)\mu m.$$

This is affine in μ and in m separately, so its minimum over $[0,1]^2$ is attained at a corner. It is positive at the corners with $\mu m = 0$, and at $\mu = m = 1$ it equals twice the left side of Lemma 27(iii).

(c) $c = -g_2$, so $A = (2pR, 2R, -(R^2+2R-1))$. Here $h_0 = -4(P-\mu)mpR$ and $h_2 = 0$, and $A_1B_1 \le 0$. If $mR \le P$, then $|B_1| \le P|A_1|$, $h_1 = 0$ and $\psi = 2P(1+\mu)(R-1)^2 + 4(P-\mu)mpR$, so $x\psi > 2T$ by (17) and Lemma 27(iv). If $mR > P$, then $R \ge 4$, since $mR \le 2 < P$ when $R = 2$; now $h_1 = 4\mu R(mR - P)$ by (10), and

$$\psi = 2P(1+\mu)(R-1)^2 + 4\mu PR + 4mR(Pp - \mu(2R+1)).$$

As $Pp - \mu(2R+1) \ge 2(R+1) - (2R+1) > 0$ and $mR > P$, we get $\psi > 2P(R-1)^2 + 4P^2(R+1) + 2\mu P(R^2 - 4R - 1)$. The right side is affine in μ, and x times it exceeds $2T$ at $\mu = 0$ by Lemma 27(v) and at $\mu = 1$ by Lemma 27(vi), since $(R-1)^2 + R^2 - 4R - 1 = 2R(R-3)$.

(d) $c = \pm g_3$. By (16) with $|B_0| = 2mpR$, $\psi \ge 2P(1+\mu)(R^2+1) - 4\mu mpR$. If $J \le b-2$, then $m \le \mu_1$ by (9), so $2mpR \le 2pR(P^2+1)/x^2 < P(R^2+1)$ by Lemma 27(vii); hence $\psi \ge 2P(R^2+1)$ and $x\psi \ge 2P(P+1)(R^2+1) > 2T$. If $J = b-1$, then $m = \mu_{-1} = 1$, $\mu = \mu_{b-2} > 0$ and $x\rho = P - \mu$, and directly

$$\psi - 2x\delta_2(p)\rho \ge 2\mu\big[(R^2+1)x - 2pR\big] \ge 2\mu(R^2 - 2R + 3) > 0.$$

(e) $c = \pm g_4$. By (16) with $|B_0| + |B_1| = 2mR(2R+1)$ and $m \le 1$, $\psi \ge 4PR^2(1+\mu) - 4\mu R(2R+1) = 4R(PR-1) + 4R\big[\mu R(P-2) + 1 - \mu\big] \ge 4R(PR-1)$, so $x\psi > 2T$ by Lemma 27(viii).

These are all the columns $c \ne -s_2$. □

Lemma 29. *Let b be even. If $c_b = g_2$, then $\sum_{j<b} \psi_j > 4Rx\rho$.*

Proof. All ψ_j are nonnegative by Lemmas 22 and 26. Here $\zeta_{b-1} = g_2$, so $B = M\zeta_{b-1} = (-2pR, -2R, R^2+2R-1)$, and $x\rho = P - \mu$ with $\mu = \mu_{b-2}$. Let $c = c_{b-1}$, $A = Mc$ and $h_i = h_\mu(A_i, B_i)$.

Suppose first that $c \ne -g_2$ or $R \ge 4$. We show that $\psi_{b-1} > 4R(P-\mu)$.

- $c = \pm g_4$: by (16), $\psi_{b-1} - 4R(P-\mu) \geq 4PR^2(1+\mu) - 4\mu R(R+2) - 4R(P-\mu) = 4PR(R-1) + 4\mu R(PR-R-1) > 0$.
- $c = \pm g_3$: by (16), $\psi_{b-1} - 4R(P-\mu) \geq 2P(1+\mu)(R^2+1) - 4\mu pR - 4R(P-\mu) = 2P(R-1)^2 + 2\mu[P(R^2+1) - 2R^2] > 0$.
- $c = g_2$ or $c = -s_2$: then $A_0 = B_0 < 0$, so $h_0 = -4(P-\mu)pR$, while $h_1, h_2 \leq 0$, for g_2 because $A_i = B_i$ and for $-s_2$ by Lemma 22. Hence $\psi_{b-1} \geq 4(P-\mu)pR > 4R(P-\mu)$.
- $c = s_2$: here $A = (2pR, -2R^2, R^2+1)$, so $h_0 = 0$, $h_1 = -4(P-\mu)R$ and $h_2 = 2\mu(R^2 + 2R - 1) - 2P(R^2+1)$ by (10). Hence $\psi_{b-1} - 4R(P-\mu) = 2P(R^2+1) - 2\mu(R^2+2R-1) \geq 2(R^2 - 2R + 3) > 0$.
- $c = -g_2$ and $R \geq 4$: by Lemma 23, $\psi_{b-1} = 2P(1+\mu)(R-1)^2$, and $\psi_{b-1} - 4R(P-\mu) \geq 2P(R^2 - 4R + 1) > 0$.

Now let $c = -g_2$ and $R = 2$, so that $p = 3$ and $P \geq 4$. By Lemma 23, $\psi_{b-1} = 2P(1 + \mu_{b-2})$ and $\zeta_{b-2} = -\mu_0 g_2$, so $B' = M\zeta_{b-2} = m\,(12, 4, -7)$ with $m = \mu_0 < 1$. We show that $\psi_{b-2} \geq 2P(1+\mu')$, where $\mu' = \mu_{b-3}$, for every column $c' = c_{b-2}$. Here $Mg_1 = (12, -8, 5)$, $Mg_2 = (-12, -4, 7)$, $Mg_3 = (0, -12, 3)$ and $Mg_4 = (0, 0, 9)$, and

$$22(P-\mu')m \geq 2P(1+\mu'), \tag{18}$$

because $x \cdot 22(P-\mu')m = 22(P-\mu')(P-1) \geq 22(P-1)^2 \geq 4P(P+1) \geq 2P(1+\mu')x$; indeed $22(P-1)^2 - 4P(P+1) = 18P^2 - 48P + 22 > 0$ for $P \geq 4$. Let h'_i be the cells of ψ_{b-2}.

- $c' = \pm g_4$: by (16), $\psi_{b-2} \geq 16P(1+\mu') - 32\mu' m \geq 2P(1+\mu') + 14P(1+\mu') - 32\mu' > 2P(1+\mu')$.
- $c' = \pm g_3$: by (16), $\psi_{b-2} \geq 10P(1+\mu') - 24\mu' m > 2P(1+\mu')$.
- $c' = \pm g_2$: $|B'_i| \leq |(Mc')_i|$ for all i, so all cells are ≤ 0 and $\psi_{b-2} \geq P(1+\mu')\Delta_2 = 2P(1+\mu')$.
- $c' = s_2$: $h'_0 = -24(P-\mu')m$ and $h'_1, h'_2 \leq 0$ by Lemma 22, so $\psi_{b-2} \geq 24(P-\mu')m$, and (18) applies.
- $c' = -s_2$: then $Mc' = (-12, 8, -5)$, $h'_0 \leq 0$ and $h'_1 = -8(P-\mu')m$. If $7m \leq 5$, then $h'_2 = -14(P-\mu')m$ and $\psi_{b-2} \geq 22(P-\mu')m$. If $7m > 5$, then $h'_2 = 14\mu' m - 10P$ by (10), and $\psi_{b-2} \geq 8(P-\mu')m - 14\mu' m + 10P \geq 8Pm - 22m + 10P > 4P \geq 2P(1+\mu')$.

Finally put $u = \mu_{b-2}$ and $v = \mu_{b-3}$, so that $xu = P - v$ and $x\rho = P - u$. By (4), $v > \mu_\infty = P/(P+2)$, since $b-3$ is odd or $v = \mu_{-1} = 1$. Using $xu = P - v$, one finds

$$x[2P(1+u) + 2P(1+v) - 8(P-u)] = 2(P-2)[(P+2)v - P] > 0.$$

Hence $\psi_{b-1} + \psi_{b-2} \geq 2P(1+u) + 2P(1+v) > 8x\rho = 4Rx\rho$. □

Proof of Theorem 25. Let $Y \in \{\pm 1\}^{3\times(b+1)}$ with $y_{2b} = -1$. By Lemmas 22 and 26, all ψ_j are nonnegative, so Lemma 21 gives

$$x^{1-b}(\Theta - G(Y)) \geq \sum_{j<b} \psi_j + x\rho(\Delta(c_b) - 2\delta_2(p)). \tag{19}$$

If $Y = Y^-$, then $G(Y) = \Theta$. Otherwise let J be the largest index with $c_J \neq (-1)^{J+1} s_2$.

If $J < b$, then $c_b = -s_2$ as b is even, so $\Delta(c_b) = 0$. By Lemma 23, $\zeta_J = \varepsilon \mu_{b-2-J} s_2$ with $\varepsilon = (-1)^J$, so $\psi_J = \psi_{\mu_{J-1}}(\varepsilon c_J, \mu_{b-2-J} s_2)$ with $\varepsilon c_J \neq -s_2$. By Lemma 28, $\psi_J > 2x\delta_2(p)\rho$, and (19) gives $G(Y) < \Theta$.

If $J = b$, then $c_b \neq -s_2$ and $y_{2b} = -1$, so c_b is g_2, g_3 or $-g_4$. If $c_b = -g_4$, then $\Delta(c_b) - 2\delta_2(p) = 2R^2 - 2 > 0$ by (14), and $G(Y) < \Theta$. If $c_b = g_2$, then $\Delta(c_b) - 2\delta_2(p) = -4R$, and $G(Y) < \Theta$ by Lemma 29. If $c_b = g_3 = g_+$, then Lemma 21 gives $x^{1-b}(\Theta - G(Y)) = \sum_j \psi_j + 2x((M\zeta_{-1})_2)^- \geq 0$. If equality holds, all ψ_j vanish, so every c_j with $j < b$ is $\pm s_2$ by Lemma 26, and $Y = Y^+$ by Lemma 24. Conversely $G(Y^+) = \Theta$ by Lemma 24. □

Remark 30. Two families have explicit gaps. For $b = 2$, $q = 3$ and $Y = s_2(1, 1, -1)^\mathsf{T}$, which is case (a) of Lemma 28 with $J = 0$, we have $\Theta - G(Y) = 4(p - 3)$ (Remark 17). For $p = 3$ and $Y = (1, 1, -1)^\mathsf{T} s_b^\mathsf{T}$, whose columns are $\pm g_2$, we have $\Theta - G(Y) = 2(d_b(q) - 5\delta_b(q))$, which is $4(q - 3)$ for $b = 2$.

12 A conditional reduction for larger exponents

This section proves no case of Conjecture 14 beyond Theorem 5. We use the notation of Section 10. For a fixed exponent $a \geq 2$ of p, and for all real p, q satisfying (8) in one of its two regions, Proposition 31 deduces the following conclusion from the conditions (S_a), (F_a) and (C_a), which do not depend on b: for all $b \geq 1$ with $b \equiv a \pmod 2$, every $Y \in \{\pm 1\}^{(a+1)\times(b+1)}$ with $y_{ab} = -1$ satisfies $G(Y) \leq \Theta$, with equality exactly for Y^- and Y^+. The proposition is conditional. We do not claim that its conditions hold for any $a \geq 3$; version 1.5 verified them for $3 \leq a \leq 8$ by computer, and that verification is not part of this version.

Let $\mathcal{R}_1$ and $\mathcal{R}_2$ be the two sets of pairs (p, q) in (8), and put $P_{\min} = 2$ on $\mathcal{R}_1$ and $P_{\min} = 4$ on $\mathcal{R}_2$. For real y let $\eta_0(y) = y$ and $\eta_{n+1}(y) = (P - \eta_n(y))/x$, so that $\mu_{j+n} = \eta_n(\mu_j)$. Put $\hat{D}_1 = d_1(x)/x = (3x-4)/x$ and $\hat{D}_2 = d_2(x)/x^2 = (5x^2-8x+4)/x^2$. For a even, let $\mathcal{F}_a$ be the set of the triples (μ, m, r) of the forms $(1, m, (P-m)/x)$ with $\mu_0 \leq m \leq \mu_\infty$; $(\mu, 1, (P-\mu)/x)$ with $\mu_0 \leq \mu \leq \mu_\infty$; (μ, m, μ) with $\mu_\infty \leq \mu \leq \mu_1$ and $\mu_0 \leq m \leq \mu_\infty$; and (μ, m, m) with $\mu_0 \leq \mu \leq \mu_\infty$ and $\mu_\infty \leq m \leq \mu_1$. For a odd, let $\mathcal{F}_a$ consist of $(1, 1, \mu_0)$; the triples $(1, m, (P-m)/x)$ and $(\mu, 1, (P-\mu)/x)$ with $\mu_\infty \leq m, \mu \leq \mu_1$; and the triples (μ, m, μ_∞) with μ, m both in $[\mu_0, \mu_\infty]$ or both in $[\mu_\infty, \mu_1]$.

Proposition 31. *Let $a \geq 2$, and suppose that the following conditions hold at every point of $\mathcal{R}_1$, or at every point of $\mathcal{R}_2$.*

(S_a) $P_{\min}\Delta(c) > \sum_{i=0}^{a} \max\{0,\ \omega_i - P_{\min}|(Mc)_i|\}$ *for every* $c \in \{\pm 1\}^{a+1}$ *other than* $\pm s_a$.

(F_a) $\psi_\mu(c, ms_a) > 2x\delta_a(p)\, r$ *for every* $c \in \{\pm 1\}^{a+1}$ *other than* $-s_a$ *and every* $(\mu, m, r) \in \mathcal{F}_a$.

(C_a) Every $g \in \{\pm 1\}^{a+1}$ *with* $g_a = -1$, *other than* $-(-1)^a s_a$ *and* g_+, *satisfies* $\Delta(g) > 2\delta_a(p)$ *or the following three conditions, where* $\gamma = 2\delta_a(p) - \Delta(g)$.

(C1) For $0 \leq k \leq 4$, *every* $y \in I_k$ *and every* $c \in \{\pm 1\}^{a+1}$ *other than* $(-1)^{k+1} g$,

$$\sum_{n=1}^{k} P(1 + \eta_n(y))\Delta(g) + \psi_y(c, (-1)^k \mu_{k-1} g) > x\, \eta_{k+1}(y)\, \gamma,$$

where $I_k = [\mu_0, \mu_\infty]$ *if* $k \equiv a \pmod 2$, *and* $I_k = [\mu_\infty, \mu_1] \cup \{1\}$ *otherwise.*

(C2) $\sum_{n=0}^{4} P(1 + \eta_n(y))\Delta(g) > x\, \eta_5(y)\, \gamma$ *for every* $y \in [\mu_0, \mu_\infty]$ *if a is even, and for every* $y \in [\mu_\infty, \mu_1]$ *if a is odd.*

(C3) $\Delta(g)(\hat{D}_1/\mu_1 + 2) > 2\delta_a(p)$ *if a is even;* $\Delta(g)\hat{D}_1/\mu_0 > 2\delta_a(p)$ *and* $\Delta(g)(\hat{D}_2/\mu_\infty + 2) > 2\delta_a(p)$ *if a is odd.*

Then at every point of that region and for every $b \geq 1$ *with* $b \equiv a \pmod 2$, *every* $Y \in \{\pm 1\}^{(a+1)\times(b+1)}$ *with* $y_{ab} = -1$ *satisfies* $G(Y) \leq \Theta$, *with equality exactly for* Y^- *and* Y^+.

Proof. First, $\psi_\mu(c,\zeta) > 0$ for all $c \neq \pm s_a$, $0 \le \mu \le 1$ and $\zeta \in [-1,1]^{a+1}$. Indeed, by Lemma 22 and $|(M\zeta)_i| \le \omega_i$, $\psi_\mu(c,\zeta)$ is at least $P(1+\mu)\Delta(c) - 2\mu\sum_i \max\{0, \omega_i - P|(Mc)_i|\}$, which is affine in μ; at $\mu = 0$ it equals $P\Delta(c) > 0$, and at $\mu = 1$ it is at least twice the difference of the two sides of (S_a), since $P \ge P_{\min}$. So, by Lemma 22, all ψ_j are nonnegative, $\psi_j = 0$ only if $c_j = \pm s_a$, and Lemma 21 gives

$$x^{1-b}(\Theta - G(Y)) \ge \sum_{j<b} \psi_j + x\rho(\Delta(c_b) - 2\delta_a(p)). \tag{20}$$

If $Y = Y^-$, then $G(Y) = \Theta$. Otherwise let J be the largest index with $c_J \neq (-1)^{J+1} s_a$.

Case $J < b$. Then $\Delta(c_b) = 0$, and as in the proof of Theorem 25, $\psi_J = \psi_\mu(\varepsilon c_J, m s_a)$ with $\mu = \mu_{J-1}$, $m = \mu_{b-2-J}$, $\varepsilon = (-1)^J$ and $\varepsilon c_J \neq -s_a$. We claim that $(\mu, m, r) \in \mathcal{F}_a$ for some $r \ge \rho$; then $\psi_J > 2x\delta_a(p)\rho$ by (F_a), and $G(Y) < \Theta$ by (20). Let a, and hence b, be even. If $J = 0$, then $\mu = \mu_{-1} = 1$, $m = \mu_{b-2} \in [\mu_0, \mu_\infty]$ by (4), and $\rho = (P-m)/x$. If $J = b-1$, then $m = 1$, $\mu = \mu_{b-2}$ and $\rho = (P-\mu)/x$. If $1 \le J \le b-2$, the indices $J-1$ and $b-2-J$ are nonnegative with odd sum; by (4), the value with the even index lies in $[\mu_0, \mu_\infty]$ and the value with the odd index lies in $[\mu_\infty, \mu_1]$ and exceeds $\rho = \mu_{b-1}$, since the μ_j with odd j decrease. Now let a be odd. If $b = 1$, then $J = 0$ and $(\mu, m, \rho) = (1, 1, \mu_0)$. If $b \ge 3$, the cases $J = 0$ and $J = b-1$ are as before, now with $\mu_{b-2} \in [\mu_\infty, \mu_1]$; and if $1 \le J \le b-2$, the indices $J-1$ and $b-2-J$ have the same parity, while $\rho = \mu_{b-1} < \mu_\infty$ as $b-1$ is even.

Case $J = b$. Then $g = c_b$ satisfies $g_a = -1$ and $g \neq -(-1)^a s_a$. If $g = g_+$, then Lemmas 21 and 24 give $x^{1-b}(\Theta - G(Y)) = \sum_j \psi_j + 2x((M\zeta_{-1})_a)^- \ge 0$; equality forces all ψ_j to vanish, so that $c_j = \pm s_a$ for $j < b$ and $Y = Y^+$ by Lemma 24, while $G(Y^+) = \Theta$. If $\Delta(g) > 2\delta_a(p)$, then $G(Y) < \Theta$ by (20). Otherwise (C1)–(C3) hold, and by (20) it suffices to show $\sum_j \psi_j > x\rho\gamma$. Let k be the largest integer with $k \le b$ and $c_{b-i} = (-1)^i g$ for $1 \le i \le k$. By Lemma 23, $\psi_{b-i} = P(1+\mu_{b-i-1})\Delta(g)$ for $1 \le i \le k$.

- If $k < b$ and $k \le 4$, put $y = \mu_{b-2-k}$, so that $\mu_{b-i-1} = \eta_{k+1-i}(y)$ and $\rho = \eta_{k+1}(y)$. Then $y \in I_k$: if $b-2-k \ge 0$, by (4), as $b-2-k \equiv a-k \pmod 2$; and if $b = k+1$, then $y = \mu_{-1} = 1$ and $k \not\equiv a \pmod 2$. The column $j = b-1-k$ satisfies $c_j \neq (-1)^{k+1} g$ and, by Lemma 23, $\zeta_j = (-1)^k \mu_{k-1} g$; so $\psi_j = \psi_y(c_j, (-1)^k \mu_{k-1} g)$, and (C1) gives $\sum_{i=1}^k \psi_{b-i} + \psi_j > x\rho\gamma$.
- If $5 \le k < b$, then $b \ge 6$, and $y = \mu_{b-6}$ lies in $[\mu_0, \mu_\infty]$ if a is even and in $[\mu_\infty, \mu_1]$ if a is odd. The columns $b-1, \dots, b-5$ give $\sum_{i=1}^5 \psi_{b-i} = \sum_{n=0}^4 P(1+\eta_n(y))\Delta(g)$, and $\rho = \eta_5(y)$, so (C2) applies.
- If $k = b$, then $\sum_j \psi_j = x\Delta(g)(\hat{D} - \rho)$, and the claim reads $\Delta(g)\hat{D}/\rho > 2\delta_a(p)$. By Remark 9, $\hat{D} = \hat{D}_1 + 2\sum_{i=1}^{b-1} \mu_i$. If b is even, then $\rho = \mu_{b-1} \le \mu_1$, so $\hat{D}/\rho \ge \hat{D}_1/\mu_1 + 2$; if $b = 1$, then $\hat{D}/\rho = \hat{D}_1/\mu_0$; and if $b \ge 3$ is odd, then $\hat{D} \ge \hat{D}_2 + 2\mu_{b-1}$ and $\rho < \mu_\infty$, so $\hat{D}/\rho \ge \hat{D}_2/\mu_\infty + 2$. Since $\Delta(g) > 0$, (C3) applies. □

13 Formal verification

All the results are formally verified in Lean 4 (version 4.34.1) with Mathlib [8] (commit `d13f23b7`). Each mathematical claim of version 2 is a statement of Table 1, a definition, a step of a written proof of one of them, or an observation that is immediate from the definitions. Statements about the literature are marked by their citations and are not formal. Where the paper gives no written proof, the result rests on the formal proof alone: Proposition 6 and the Ramanujan-sum formula and row sums of Section 2, which it cites from [6]; and, in Remark 16, the energies of the graphs for $n = pq$, the identification of these graphs, the values 512 and 1472 and the fact that the maximisers are not multiplicative.

Statements before formal proofs. On 27 September 2026 a Claude Code agent (model Claude Opus 5.5) wrote a first Lean statement file from version 1.5, with known-answer tests of its definitions, and an independent Claude Code agent (same model) reviewed it. The second version of the file, with 33 statements and 84 known-answer tests, was reviewed by a Codex agent that had not written it, and it was frozen on 27 September 2026 at 21:35 UTC. By 28 September 2026 Codex had proved 29 of them. The other four were `Theorem5` ($\min\{a,b\} \le 8$), `Proposition15`, `Proposition31` and `Certificates` (that the conditions of Proposition 31 hold for $3 \le a \le 8$). That file and the record of the 29 accepted statements are kept.

On 6 November 2026 the author decided to narrow the paper to $\min\{a,b\} \le 2$ rather than formalise the certificates. OpenAI Codex then wrote the present statement files. Compared with the frozen file, apart from comments, they replace `Theorem5` by `Theorem5MinTwo` and delete `Proposition15` and `Certificates`; every other definition and statement is unchanged, including `Proposition31` for general a. They add 20 statements on details of the text. Two Codex agents that had not written the files reviewed the statements and the scope, a third review followed after the files had been compiled, and the 51 statements were frozen on 7 November 2026 at 03:05 UTC, with the 84 tests (their statements word for word; six lines of their proofs were changed to remove warnings) and 10 regression tests. The order was not strict: the proofs of the 29 statements, and the older Lean modules they use, were written before these files, and the 20 added statements were written after those proofs.

Formal proofs. The proofs are in files that import the statement files and do not modify them. The modules `Research/ICG*` go back to the Lean formalisation of the earlier versions of this paper, written, according to the disclosure of those versions, by Claude Code agents. On 27 and 28 September 2026 Codex proved 29 of the 33 statements in further modules that use them. On 6 and 7 November 2026 Codex moved these proofs to Lean modules under `Research/Backfill/Paper1/`, with small compatibility changes (import paths and the names of deprecated lemmas), and proved `Theorem5MinTwo`, the 20 added statements and `Proposition31`. For Proposition 31 it started from drafts written in September 2026. A review of the coverage of the text then found that two formulas of written proofs had no named Lean statement: the path, signs and last entry for Y^+ in the proof of Lemma 24, and the monotonicity of f in the proof of Lemma 27. Four lemmas (`plus_zeta_closed`, `plus_signed_cell_closed`, `plus_terminal_closed` and `ratio_monotonicity`) were added for them; their statements were reviewed before they were proved. They are not counted among the 51 statements.

What is proved. The module `FinalContractAudit` states each of the 51 statements of Table 1 with its exact frozen type, proves it by the theorem `CodexPaper1.check_X` of the proof files, where `X` is the name of the statement, and prints its axioms. It does the same for the four lemmas above and for the calibration $\mathcal{E}(\mathrm{ICG}_3(\{1\})) = 4$, which is computed from the actual graph and checks that the formal energy is not a trivial function.

Checks. The Lean package of version 2 has 70 modules: the statement and test files, the proofs and the final audit. On 7 November 2026 (04:50–05:21 UTC) it was built from source in two new empty directories, one after the other, on the author's Windows computer. Each module was compiled and checked with `leanchecker`. All 280 commands, 140 in each directory, ended with exit code 0 on their first attempt, with no error, no warning and no `sorry`. The 70 compiled `.olean` files of the two directories are byte for byte identical. Mathlib and the toolchain were not rebuilt. In each directory, 200 axiom listings printed by `#print axioms` cover 144 distinct declarations, and each consists of exactly `propext`, `Classical.choice` and `Quot.sound`. A static scan of the sources, outside comments, found no `sorry`, `admit`, `native_decide`, `bv_decide`, `implemented_by`, `extern` or `unsafe`, no `axiom` declaration and no import of the deleted statements. Two modules keep three local options, all from the proofs of

In the paper	Lean statement
Theorems 1, 2	`Theorem1`, `Theorem2`
Section 1: opposite parity, all distinct odd primes	`SummaryOppositeParity`
Theorems 3, 4	`Theorem3`, `Theorem4`
Theorem 5	`Theorem5MinTwo`
Section 1: every order with $\min\{a, b\} \leq 2$	`CompleteMinTwo`
Section 2: Ramanujan sums; symmetry and row sums	`RamanujanEntry`, `TransformRows`
Proposition 6	`Proposition6`
Section 3: the path is linear and convex	`PathLinear`, `PathConvex`
Lemma 7 and its proof	`Lemma7`, `PathCoordinates`
Section 4 and (9): bounds on μ_j	`MuBounds`
Lemma 8	`Lemma8`
Remark 9	`Remark9`, `AlternatingPath`
Lemmas 10, 11	`Lemma10`, `Lemma11`
Theorem 12, Remark 13	`Theorem12`, `Remark13`
Section 8: formula (2)	`GeneralGNorm`, `GeneralEqG`
Section 8: the value of the first set; the bound is not attained	`Section8OddSet`
Remark 16	`Remark16`, `PQGraphModels`, `MultiplicativeBridge`, `MultiplicativeMaxComparison`
Remark 17	`Remark17`, `NearExampleDetails`
Sections 9, 10: (3), (8) and consequences	`ParameterRegions`
Section 9: (4)	`MuClosedParity` (and `MuBounds` above)
Lemmas 18, 19, 20	`Lemma18`, `Lemma19`, `Lemma20`
Sections 9, 10: values of h_μ, (10), (11)	`CellValues`, `CellStrictBoundary`
Section 10: (12), (13)	`MatrixFacts`, `MatrixBounds`
Lemmas 21, 22, 23, 24	`Lemma21`, `Lemma22`, `Lemma23`, `Lemma24`
Theorem 25	`Theorem25`
Lemmas 26, 27, 28, 29	`Lemma26`, `Lemma27`, `Lemma28`, `Lemma29`
Remark 30	`Remark30`
Proposition 31	`Proposition31`

Table 1: The 51 statements of version 2. The 29 whose names begin with `Theorem`, `Lemma`, `Remark` or `Proposition`, together with `SummaryOppositeParity` and `Section8OddSet` (31 in all), are in the namespace `BackfillPaper1.Challenge`; the other 20 are in `BackfillPaper1.NarrowSupplement`.

September 2026: `linter.unnecessarySeqFocus false` once in `ICGEqualParityBGraph` and `maxHeartbeats 2000000` twice in `RemarkPQFinite`. They limit how long Lean may work or what it warns about, and they do not change what the kernel accepts.

After both builds had ended, two Codex agents that had not written the current package reviewed it; neither ran Lean. One recomputed the hashes of the 656 files of the evidence archive, including the 568 build artefacts, read all 280 logs and approved the machine evidence (7 November 2026, 05:28 UTC). The other read the statements and the central proofs, compared the statements with the frozen types and the text of version 1.5, checked a map from 15 groups of paragraphs of the written proofs to 100 named Lean declarations, and approved the mathematics, the statements and this coverage (05:34 UTC). Both had taken part in read-only reviews of earlier stages, and the older proofs contain earlier work by Codex, so these reviews are not a review by another model family.

In all, on 6 and 7 November 2026 the statement files, the development of the proofs, an earlier build of 68 modules in two new directories (272 commands) and the final build ended 713 compile and check commands, of which 690 succeeded and 23 failed, counting warnings as failures; the 161 commands outside the two builds had 138 successes and 23 failures. Their logs are kept. One early run of the statement files on Windows ended with a console interrupt, for a reason we did not establish. On 7 November 2026 (06:10–17:27 UTC) Claude Code rebuilt the

whole Lean project of the repository with these 70 modules on the author's Windows computer, from an empty directory, one module at a time and not with `lake`: all 2954 modules compiled, and 2953 passed `leanchecker`. The remaining module is the root of the project, which has only imports and no constants, so it was not replayed. The computer ran out of memory during the first run: some modules of Paper 7 of the repository could not read compiled files, and they and the modules importing them were run again, with fewer modules at a time, in two further runs. No run ended in a Lean error, and no module of this paper was among them. The 56 axiom lists printed by the final audit (the 51 statements, one calibration and four internal lemmas) consist of exactly the three axioms above. This paper has not been checked with comparator or with an independent kernel. The statement files, the tests, the proofs and the final audit are in version 1.16.0 of the repository [1], directory `lean/Research/Backfill/Paper1/` and the modules `lean/Research/ICG*`; the receipts of the builds are in `certificates/icg-q3-general/lean-package-v2/` and `logs/clean-replay-v1.16.0-receipt.json`.

Definitions the reader must trust. From Mathlib the statements use finite simple graphs and their real adjacency matrices, the eigenvalues of a Hermitian matrix given by the spectral theorem, graph isomorphisms and the complete graph, subtraction on `Fin n`, gcd, proper divisors, divisors, primes, Euler's function, the real cosine and π, and finite sets and sums. The statement files define: the relation $\gcd(u-v,n)\in\mathcal{D}$ on `Fin n` and the graph it generates (`SimpleGraph.fromRel` adds $u\neq v$ and symmetrises); the energy as the sum of the absolute values of the eigenvalues; admissible sets (nonempty sets of proper divisors, `IsConnSet`, which does not ask the graph to be connected); `MaxEnergy` (all the listed sets are admissible, every admissible set has energy at most the value, with equality exactly for the listed sets); the sets `evenSet` and `oddSet`; distinct odd primes; the closed forms d_k (`dClosed`) and δ_k (`deltaK`), and the norm $\|T_k(x)s_k\|_1$ (`dNorm`); $T_k(x)$, vectors and matrices as functions on $\mathbb{N}$ of which only the coordinates up to k, a and b are used; $\|\cdot\|_1$, s_k, the triple product $T_a(p)YT_b(q)^{\mathsf{T}}$ and the divisor matrix; the path z_j and the potential μ_j, with integer indices from -1 on, c_j, NA and h; G, Θ, Y^-, Y^+ and the regions of (8); Δ, ω_i, ψ_μ, ζ_j, ψ_j, g_+ and t^-; μ_∞, η_n, $\mathcal{F}_a$, I_k, $\hat{D}_1$, $\hat{D}_2$ and the conditions (S_a), (F_a), (C_a) of Section 12; the finite cosine sum for $c_m(u)$ (`ramanujanReal`); multiplicative sets and $\tilde{E}_{\max}$; the parameter ranges of Section 9; the vectors g_2, g_3, g_4; and the three graphs on `Fin p` $\times$ `Fin q` of `PQGraphModels`. They also define the simple predicates `IsSignVec`, `IsPmAlt`, `IsSignMat`, `IsPmCheckerboard`, `IsPm3`, `MatEq` and `col`, and name the statements `EqualParityClaim`, `CondAll`, `SignConclusion` and `Conjecture14`, the last of which is not claimed. The 84 known-answer tests check these definitions on examples.

Faithfulness and non-claims. Some statements are more general than the text: `Proposition6` holds for all distinct primes and all sets of proper divisors, and `Lemma18` and `Lemma19` hold for all real x above 1 and 2, respectively, and for all $m\geq 1$. `Conjecture14` is only a definition in the statement file; nothing proves or assumes it. `Proposition31` is the implication of Proposition 31, and no Lean statement says that its conditions hold for any a. No Lean statement depends on [6], [7], [3], [4] or [5], and attributions to them occur only in comments of the statement file. The attributions, and the general results of Le and Sander for arbitrary n, are not formalised, and the comparison `MultiplicativeMaxComparison` is only for orders p^aq^b. The statements say nothing about priority or about the searches of Section 1. The agreement of the Lean definitions with those of the paper was checked by the reviewing agents, not by a machine. Formal proofs may follow other routes than the written ones.

Changes in version 2. The title and the numbers of the remaining statements are unchanged. Apart from the footnote, the abstract, this section, the disclosure and the dates, version 2 differs from version 1.5 mainly as follows. Theorem 5 is stated for $\min\{a,b\}\leq 2$. Section 1 no longer says that Conjecture 14 is open for $\min\{a,b\}\geq 9$, and it reports the later work [7, 15]

and the revision of [6]; the conjecture is kept only as a historical statement. In Remark 9 the polynomial form of δ_k is added, and in Remark 16 the definition of multiplicative sets for $n = p^a q^b$; the restatement of the general results of Le and Sander and the words "by direct computation" are removed. In Remark 30 the opening and the last sentence, which compared the two differences with Θ without a precise bound, are removed; its exact formulas are kept. Section 12 now contains only Proposition 31 and its proof, which are unchanged, with a new heading and introduction. Deleted for lack of a formal proof: the case $3 \le \min\{a, b\} \le 8$ of Theorem 5; Proposition 15; the certificates of Section 12 with all their counts, programs, negative controls and checks; and the list of cross-checks of the section on verification with its paragraph on certificates, including the exhaustive searches, the tests of lemmas, the comparison with floating-point spectra and the 325 inequalities of an earlier proof of Theorem 25. Also deleted is the paragraph after Lemma 27 that described its Lean proof. The programs and logs of those computations remain in the earlier releases of [1]; they are not evidence for this version. Version 1.5 said that the closed forms of d_k and δ_k were not formalised; they now are (`Remark9`), and the statements of the main theorems use them.

Tool and computational resource disclosure

Versions 1.1 to 1.5 of this paper were produced on 25 and 26 September 2026 in a workflow directed by the author, using the AI coding agents Anthropic Claude Code (model Claude Opus 5.5) and OpenAI Codex (desktop application; models `gpt-6-astra` and `gpt-6-sol`). The author chose the research programme and directed the agents.

- A Claude Code agent found the proof of Theorem 12 and of Theorem 2, formulated Conjecture 14, and wrote verification programs.
- Another Claude Code agent wrote the Lean formalisation of those versions.
- The proof was checked line by line by a second Claude Code agent and by Codex, and the agents searched the literature.
- The results on exponents of equal parity in version 1.2, Theorem 3 and the now withdrawn Proposition 15, were found by a Claude Code research agent, and Theorem 3 was checked by an independent Claude referee agent.
- The material added in version 1.4, namely Theorems 4, 5 (then for $\min\{a, b\} \le 4$) and 25 and Sections 10–12, was found, partly formalised in Lean and drafted by Claude Code agents, and earlier write-ups of the new proofs were checked by two independent Claude Code referee agents.
- Version 1.5 extended Theorem 5 to $\min\{a, b\} \le 8$ by certification runs made by Claude Code agents, and a third Claude Code referee agent checked the extension. Version 2 withdraws this extension.

For versions 1.4 and 1.5 the author read the paper and its changes and, using a side-by-side table, compared the statements of the Lean theorems of that time with the theorems stated there.

Version 2 was produced from 27 September 2026 to 8 November 2026. On 27 September 2026 a Claude Code agent (model Claude Opus 5.5) wrote the first statement file of Section 13, and an independent Claude Code agent (same model) reviewed it. OpenAI Codex (desktop application) reviewed its second version, wrote the Lean proofs of 29 statements on 27 and 28 September 2026 and, on 6 and 7 November 2026, the present statement files, the remaining proofs and the final audit, and had them reviewed as described in Section 13. Codex's session records name the

model `gpt-6-astra` for its work of 27 and 28 September 2026 and `gpt-6.1-sol` for that of 6 and 7 November 2026; they do not name a separate model for the Codex review agents. Claude Code (model Claude Opus 5.5) compared version 1.5 claim by claim with the statements and drafted the text of version 2. On 7 November 2026 Claude Code rebuilt the whole Lean project of the repository on the author's Windows computer, and a Claude Code agent searched the literature again. An independent Claude Code agent (model Claude Fable 5.1), which had written none of the files, reviewed a draft of this paper against the statements and the evidence on 7 November 2026, and before the release a second such agent (same model) reviewed the paper, the Lean package and the receipts; neither ran Lean. The corrections of both reviews were applied. Throughout, "we" refers to this workflow; the reading and searching described in the paper were done by the agents. The author is not a professional mathematician. The author has read version 2 of this paper and, using a side-by-side table, compared the statements of the Lean theorems with the results stated here. No human mathematician has reviewed this paper. The correctness of the results rests on the Lean proofs of the statements in Table 1 and on the agreement of those statements with the paper, which was checked by the reviewing agents; the written proofs are given for the reader. AI systems are not authors of this paper; the author takes full responsibility for its content.

References

[1] H. Dong, *ai4math-results*, repository, https://github.com/Horace-Maxwell/ai4math-results, versions 1.1.0, 1.2.0, 1.4.0, 1.5.0 and 1.16.0 (2026); archived at Zenodo under the concept DOI https://doi.org/10.5281/zenodo.22970254.

[2] I. Gutman, The energy of a graph, *Ber. Math.-Statist. Sekt. Forsch. Graz* **103** (1978), 1–22.

[3] A. Ilić, The energy of unitary Cayley graphs, *Linear Algebra Appl.* **431** (2009), 1881–1889.

[4] A. Ilić and M. Bašić, New results on the energy of integral circulant graphs, *Appl. Math. Comput.* **218** (2011), 3470–3482.

[5] T. A. Le and J. W. Sander, Extremal energies of integral circulant graphs via multiplicativity, *Linear Algebra Appl.* **437** (2012), 1408–1421, https://doi.org/10.1016/j.laa.2012.04.012.

[6] J. Jiang and C. Yang, Energy maximisation for integral circulant graphs with opposite-parity exponents, arXiv:2608.29523v1, 30 August 2026; revised as arXiv:2608.29523v2, 30 October 2026, with the title *Ramanujan transforms and maximum energy of odd-total-exponent integral circulants*. The numbers of theorems, remarks and equations cited in this paper are those of v1.

[7] J. Jiang and C. Yang, Maximum energy of integral circulant graphs with same-parity exponents, arXiv:2610.36883v1, 27 October 2026.

[8] The mathlib Community, The Lean mathematical library, in *CPP 2020*, ACM, 2020, https://doi.org/10.1145/3372885.3373824.

[9] S. Park, *Exact energy maximisation for integral circulant graphs of order* p^2q^3, manuscript posted 21 August 2026, https://github.com/coshaman/papers/tree/2f730393e3d5bed4d9641d7d79b1bdc57fbff05d.

[10] D. G. Roldán, Graph energy maximisation for integral circulant graphs of order $n = p^2q^3$, arXiv:2604.09491v1, 10 April 2026.

[11] J. W. Sander and T. Sander, The exact maximal energy of integral circulant graphs with prime power order, *Contrib. Discrete Math.* **8** (2013), 19–40.

[12] J. A. Schreib, *The Roldan universality conjecture*, repository, 5 September 2026, https://github.com/jamesschreib/roldan-universality-conjecture/tree/95bb002688c65948a15cc9380e8e1f08f8bd6801.

[13] W. So, Integral circulant graphs, *Discrete Math.* **306** (2006), 153–158.

[14] The Omega Institute, *Roldán integral-circulant energy: exceptional $q = 3$ case*, issue #8338 of the `trureturing` repository, opened 16 September 2026, https://github.com/the-omega-institute/trureturing/issues/8338.

[15] The Omega Institute, *Prove Roldán Conjecture 1.3 for integral circulant energy at order p^2q^3*, pull request #19390 of the `trureturing` repository, merged 20 October 2026, merge commit `210ccfe5`, https://github.com/the-omega-institute/trureturing/pull/19390.